



Cisco Meeting Server (CMS) TAC Training



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Experience:

GC-COL TAC 2014-2022
Endpoints, VCS/Expressway,
MCU, TMS, CMS, CUCM, UCCX,
CUC, GW

Certificate:

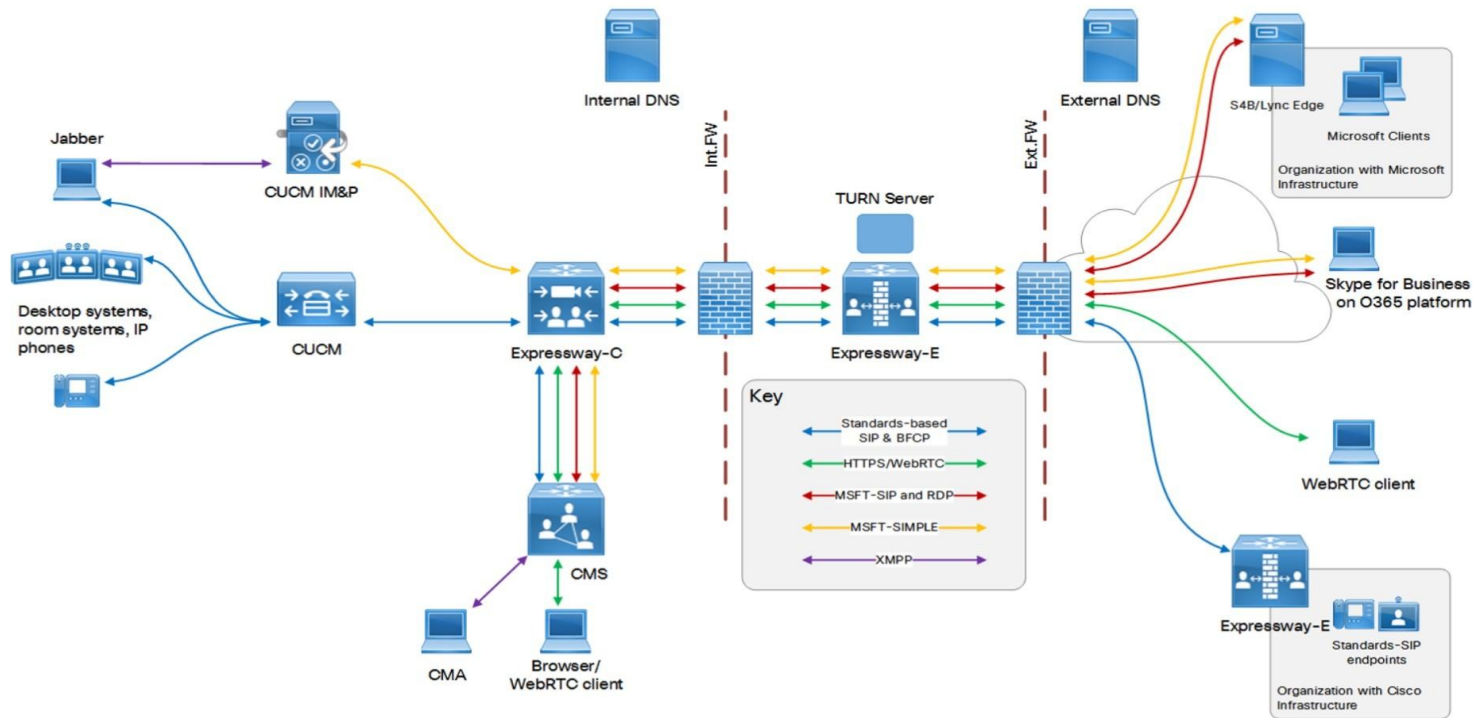


Agenda

- Cisco Collaboration Architecture
- Cisco Meeting Server Overview
- CMS Platform Options
- Deployment models and configuration
- CMS Certificates
- CMS – UC Integration (CUCM, VCS)
- CMS Dial Plan
- API, CDR, License Overview
- Callbridge Grouping
- Skype integration
- Troubleshooting and Logs

Cisco Collaboration Architecture

Preferred Call Control Solution Architecture

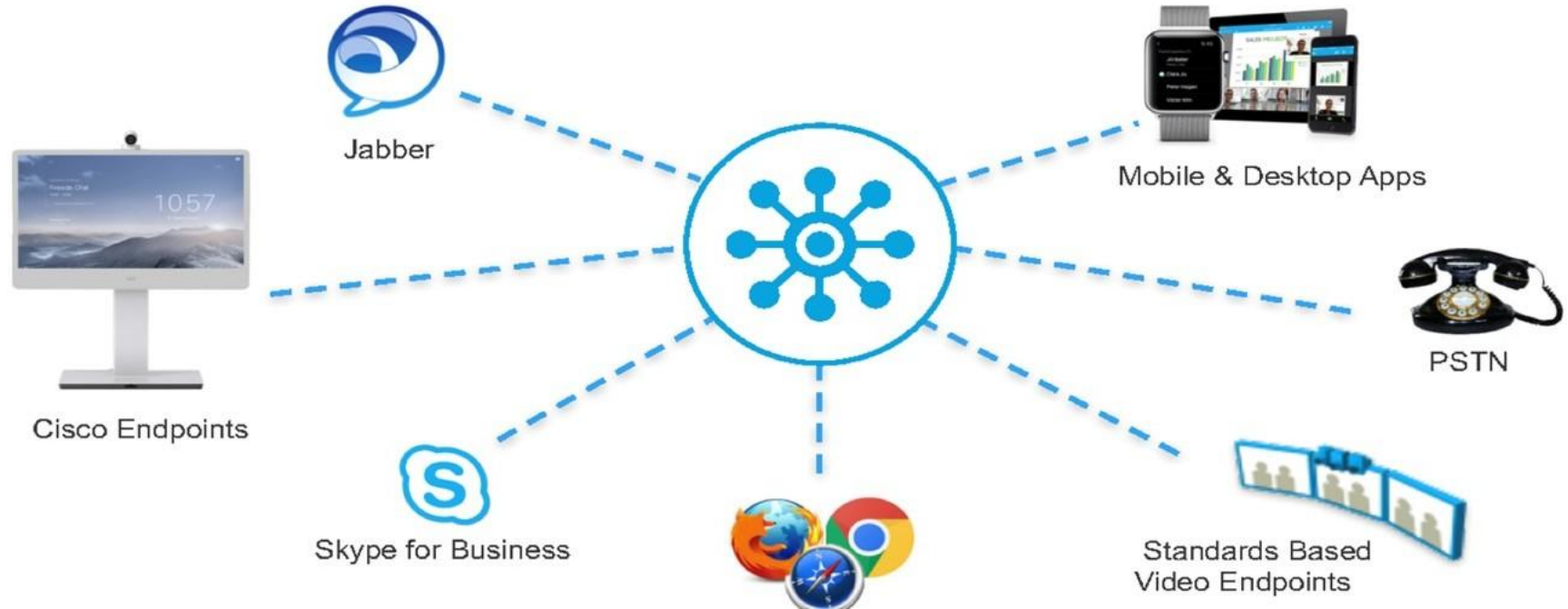


Cisco Meeting Server Overview

Cisco Meeting server (CMS)



Cisco Meeting server (CMS) - Endpoints



Cisco Meeting server (CMS) – Meetings

- **Personal Spaces:**

- Invite others to your personal space using your own join details
- With Spaces – users are in control

- **Scheduled Spaces:**

- Leverage Cisco TelePresence Management Suite (including Microsoft Outlook integration)
- One-Button-to-Push support

- **Ad-hoc with UCM:**

- Easily escalate your 1:1 calls to include more people

- **Interoperability Gateway:**

- Enable native calling and content sharing between Skype for Business



Cisco Meeting server (CMS)

- Cisco Meeting Server brings video, audio, and web communication together to meet the collaboration needs of the modern workplace. It allows anyone to create and join meetings easily, from a room or desktop video system, mobile client, or browser.
- Everyone gets a consistent, familiar meeting experience, whether they're joining a meeting using a Cisco or third-party video endpoint, a Cisco Jabber® client, Cisco Meeting App, a WebRTC compatible browser, or Skype for Business.
- Meetings can scale easily, allowing any number of people to join, and bandwidth usage is optimized in order to reduce cost. Cisco Meeting Server provides a user experience that is simple and intuitive.

CMS – Platform Options

Deployments – Platform Options



X series Server [EOS]

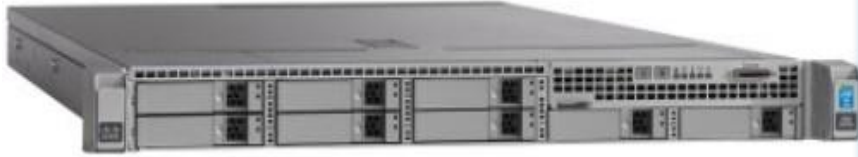


CMS 1000/2000
MM410v, MM400v (legacy)



Spec Based Server

Cisco Meeting Server 1000



- UCS C series server (1 RU)
- Pre installed CMS Software
- VMWare ESXi6.0 and above
- 70 Hyperthreaded cores
- Co-residency not supported
- CMS 1000 and Appliances can be combined in a redundant deployment, however a backup from one cannot be restored on another

CMS Virtualized Solution

Type of Calls	Cisco meeting Server 1000	Cisco Multiparty Media 410	Cisco Multiparty Media 400
HD Calls	96	64	32
SD Calls	192	128	72
Audio Calls	3000	2000	1000

Cisco Meeting Server 2000

High-Capacity Platform



- 6RU UCS
 - Based on UCS 5108 and B200 Blades
- High Capacity
 - Up to 700 HD calls (500 HD calls prior 2.4)
- Bare Metal platform

Cisco Meeting Server 2000

- Deployment Options

CMS 2000 is an Appliance (no vmware)

Cannot buy with single blade

Does not need certified UCS Engineer to deploy

All 8 blades act as single unit

Does not use clustering or cascading to achieve this

Can be clustered with another CMS2000, CMS1000 or spec based server deployment



700 HD calls



700 x N HD calls

Maximum 4000 HD calls



700 HD calls

X 8



CMS 2000



CMS 1000



UCS

CMS Spec Based VM Requirements

	Minimum	Recommended
Server manufacturer	Any	Any
Processor type	Intel Nehalem microarchitecture AMD Bulldozer microarchitecture	Intel Xeon 2600 v2 or newer
Processor frequency	2.0GHz	2.5Ghz
RAM	1GB per core	1GB per core
Storage	100GB	100GB
Hypervisor	VMWare ESXi 5.0 Update 3 Hyper-V 2012	VMWare ESXi 5.1 Update 2, 5.5 Update 1 Hyper-V 2012, 2012 R2

Reference: https://www.cisco.com/c/dam/en/us/td/docs/voice_ip_comm/uc_system/virtualization/virtualization-cisco-meeting-server.html

CMS VM Solution Recommended Setups

720p30 call legs	CPU configuration	RAM configuration	Example systems
50	Dual Intel E5-2680v2	32 GB (8x4GB)	Cisco UCS C220 M3 Dell R620 HP DL380p Gen8
40	Dual Intel E5-2650v2	32 GB (8x4GB)	Cisco UCS C220 M3 Dell R620 HP DL380p Gen8
25	Single Intel E5-2680v2	16 GB (4x4GB)	Cisco UCS C220 M3 Dell R620 HP DL380p Gen8
15	Single Intel E5-2640v2	8 GB (4x2GB)	Cisco UCS C220 M3 Dell R620 HP DL380p Gen8

Server Components

Cisco Meeting Server components

Core Components

Call Bridge	Is the component that bridges conference connections together into a single conference
WebBridge	If you are using the CMA WebRTC Client you will need to enable and configure the WebBridge
Database	The CallBridge reads from and writes to the database storing the space and configuration information
WebAdmin	The WebAdmin is a web interface to administer the CallBridge, typically running on 443, unless the WebBridge is running on the same interface, then it should be moved to 445 or 8443
Recorder	Enables automatic recording on meeting start, triggered via DTMF or administrator defined.

Cisco Meeting Server components

Core Components – continued..

Streamer

Streamer destination Url defined by API. VBrick supported as external streaming server.

H.323 GW

H.323 Gateway enables a H.323 call to connect to the CMS CallBridge

Edge Components

TURN Server

The TURN server provides firewall traversal technology, allowing the Meeting Server to be deployed behind a Firewall or NAT

SIP Edge

To support traversal of local firewalls for SIP endpoints and open and direct federation for O365, SfB and Lync calls

Deployment Models

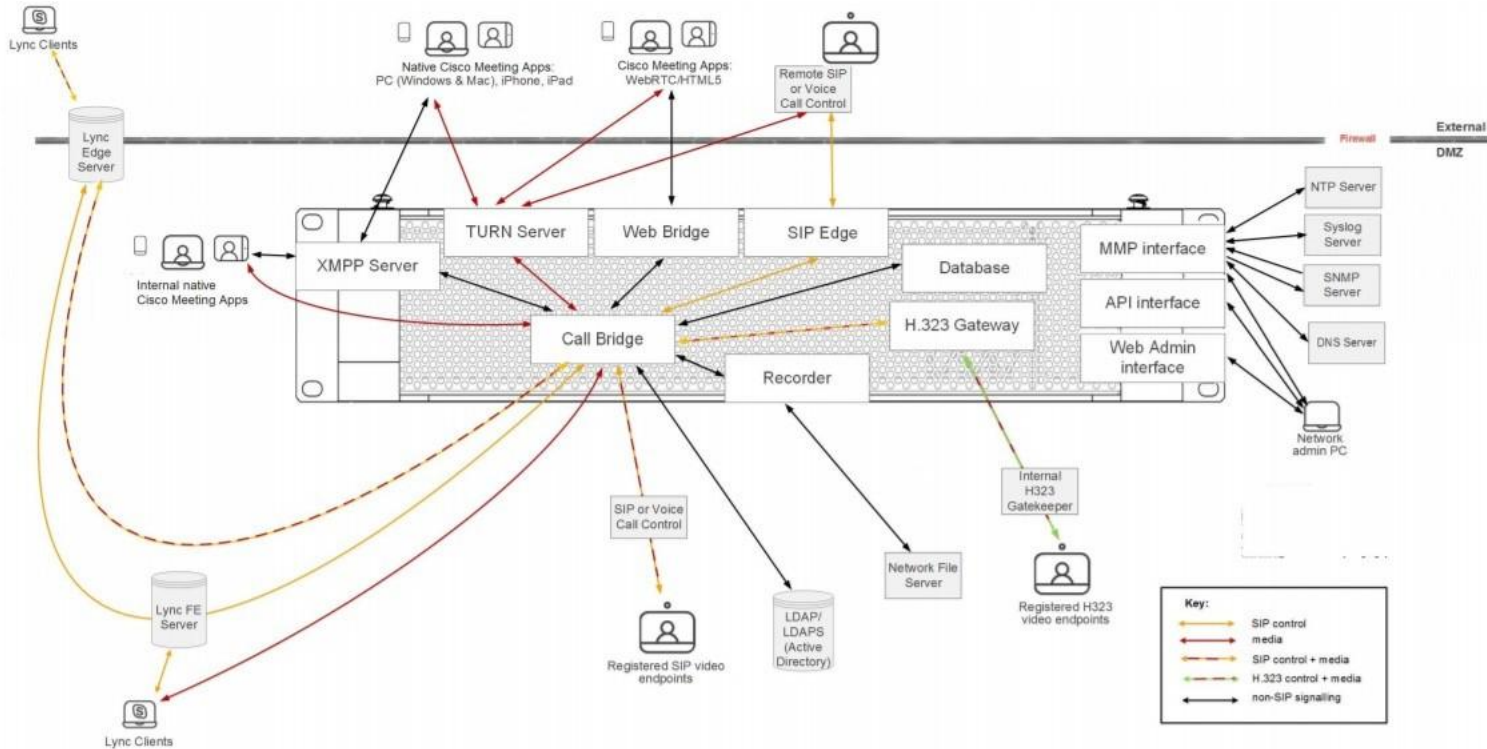
Deployment Models

Three types of deployment Models

- Single combined server
- Single Split server
- Scalable and Resilient

Single Combined Server Deployment

Single Combined Server Deployment



Server Components configuration

Web Admin

The first Component we configure is the Webadmin. The Webadmin is the Service to enable Web GUI for Meeting server

- Webadmin is specifically to configure how the Call Bridge talks to other components
- Required for Call routing configuration
- Required for Clustering configuration
- Required for viewing logs via web and increasing log levels

Web Admin Config

Configure Webadmin using MMP, We need to set Webadmin listen interface, add certificate and key, and enable the service

- “webadmin listen <iface> <port>”
- “webadmin certs <key> <cert>”
- “webadmin http-redirect enable”
- “webadmin enable”

```
Configure webadmin

Usage:
  webadmin
  webadmin restart
  webadmin enable
  webadmin disable
  webadmin listen <interface> [<port>]
  webadmin certs <key-file> <cert-file> [<cert-bundle>]
  webadmin certs none
  webadmin http-redirect <enable/disable>
  webadmin status

Core1>
```

Call Bridge

The next Component is the Callbridge. The call Bridge is the service on the Meeting Server that bridges the conference connections, enabling multiple participants to join meetings hosted on the Meeting Server

- Primary component of the solution
- Must exist somewhere in all deployments
- Media processing engine
- API integration point
- Call processing and routing component
- Fully brand-able audio and video prompts
- Supports clustering for distributed calls

Call Bridge Config

Configure a Callbridge listen interface, add certificate and key, and restart the service

- “callbridge listen <iface>”
- “callbridge certs <key> <cert>”
- “callbridge trust c2w <bundle>”
- “callbridge restart”

```
acano> callbridge
Beta options
Listening interfaces : a
Preferred interface  : none
Key file             : cms.key
Certificate file     : cms.cer
Address              : none
CA Bundle file       : root.cer
C2W trusted certs    : cms.cer
Callbridge cluster trusted certs : none
```

Web Bridge

The Webbridge service to enable WebRTC app. The WebRTC app works on HTML5-compliance browsers and uses the WebRTC standard for video and audio

- It equivalent of VCS Jabber Guest solution
- Allows for a guest to join via special link or full access to “web” version of desktop client
- Based around Websockets and WebRTC
- Chrome, Firefox, and Opera supported, Chrome is preferred
- Must use different port or IP from Web Admin if on the same server

Web Bridge Config

Set Webbridge with a listen interface, key and certificate, add a trust towards the callbridge, and enable the service

- “webbridge3 listen <iface> <port>”
- “webbridge3 certs <key> <cert>”
- “webbridge3 trust <callbridge_cert/ca_cert>”
- “webbridge3 c2w listen <interface:port>”
- “webbridge3 c2w certs <key file> <crt-fullchain-file>”
- “webbridge3 http-redirect enable”
- “webbridge3 enable”

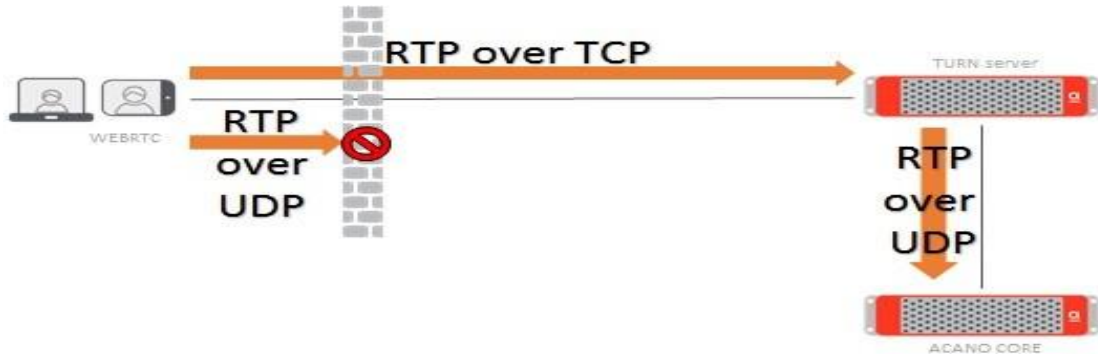
```
acano> webbridge3
Enabled                                     : true
HTTPS listening ports and interfaces       : a:443
HTTPS Key file                             : cms.key
HTTPS Full chain certificate file          : cms.cer
HTTPS Frame-Ancestors                     : none
HTTP redirect                             : Enabled, Port:80
C2W listening ports and interfaces         : a:9999
C2W Key file                              : cms.key
C2W Full chain certificate file            : cms.cer
C2W Trust bundle                          : cms.cer
Beta options                              : none
acano> █
```

TURN Server

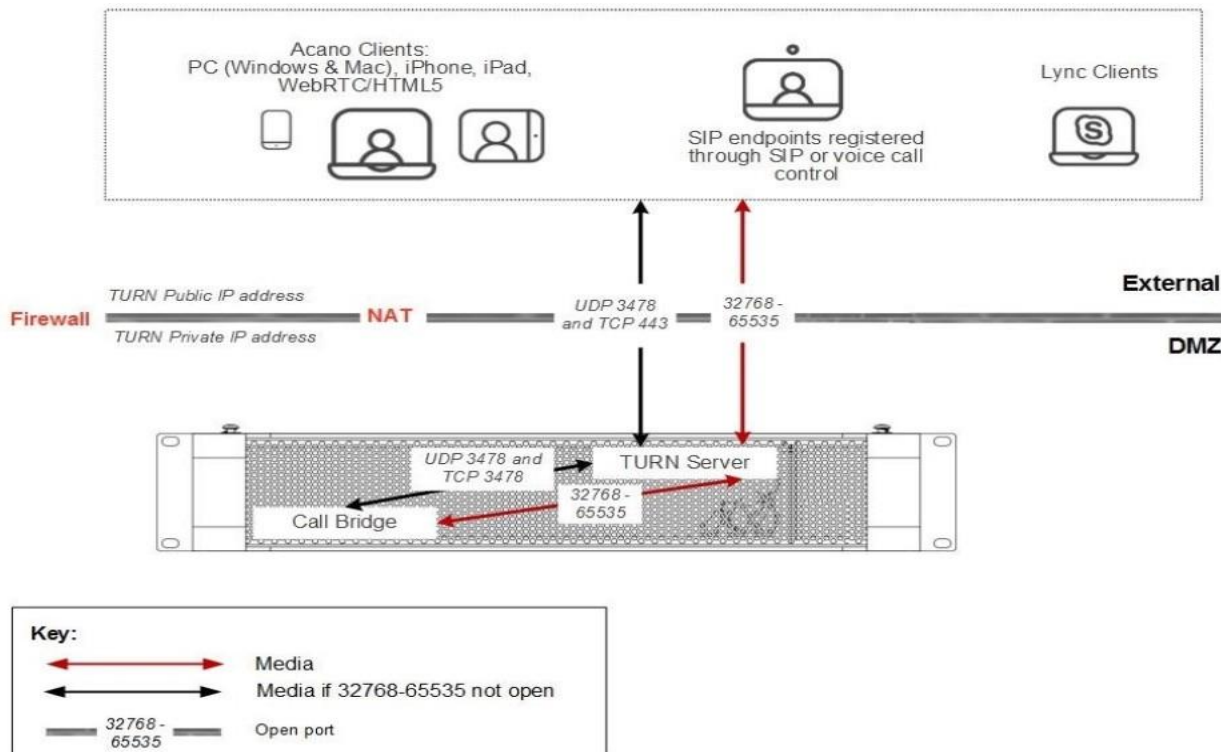
- Necessary for NAT traversal when clients are connecting externally
- Provides a media path in situations when direct media is not possible
- H.323 Capable
- Included as an option in all deployments (no additional licensing)

TURN server support for TCP to UDP interworking

- Allows TCP media from browser clients to be received
- TURN server converts this back to UDP media
- Useful when UDP traffic from browsers is blocked



TURN Server



TURN Server Config

- Basic TURN Server setup
- “turn listen <iface>”
- “turn credentials <username> <password> <domain>”
- “turn enable”
- “turn public-ip <IP>”
- “turn tls 443”

To run Turn and Webbridge on port 443 requires them to be run on separate servers/VMs, or if on the same server/VM they need to be on different interfaces and different subnets.

For maximum connectivity from external locations, Cisco recommends that port 443 is used for both the Web Bridge and

TURN Server.

Configure TURN server

Usage:

```
turn enable
turn disable
turn restart
turn credentials <username> <password> <realm>
turn public-ip <ip address>
turn del public-ip
turn listen <interface whitelist>
turn tls <port|none>
turn certs <key-file> <crt-file> [<crt-bundle>]
turn certs <none>
```

Core1> █

TURN Server settings

TURN Server address (server)	192.168.10.22
TURN Server address (clients)	5.10.20.99
Username	myusername
Password	*****
Confirm password	*****

Database

- Automatically created when a Call Bridge is created
- Stores information regarding spaces and their content
- Supports clustering if desired for redundancy
- When clustered, automatically elects a master which all Call Bridge servers will talk to
- If 5 consecutive keepalives fail, new master is elected
- A Call Bridge cluster requires a Database cluster
- Database failover can take approx. 1 minute to become operational
- Inter-database communication handled via SSL

Certificate Requirements

- Each service to be activated must have its own signed certificate and key pair (part of the configuration process).
- These certificates can be the same across multiple services.
- CMS server provides the option to generate self signed certificates with the “pki selfsigned” command along with CSRs.
- Certificates can be inspected directly via the MMP with the “pki inspect” command for troubleshooting.

Usage:

```
pki
pki list
pki inspect <file>
pki match <key> <certificate>
pki verify <cert> <CA file/cert bundle> [<CA file>]
pki unlock <key>
pki csr <key/cert basename> [<attribute>:<value>]
pki selfsigned <key/cert basename>
pki pkcs12-to-ssh <username>
```

Self signed certificates

pki selfsigned <key/cert basename>

```
CMS> pki selfsigned webadmin
```

```
.....+++
```

```
.....+++
```

Created key file webadmin.key and selfsigned certificate webadmin.crt

```
CMS> pki list           (certificates are in server root)
```

```
webadmin.key
```

```
webadmin.crt
```

CA signed Certificates

To generate the private key and Certificate Signing Request file:

- Type the “**pki csr**” command using this syntax
- **pki csr** <Key/Cert basename> <CN:value> [OU:<value>] [O:<value>] [ST:<value>] [C:<value>]
[subjectAltName:<value>]
- <Key/Cert basename>: is a string identifying the new key and CSR. Can contain alphanumeric, hyphen or underscore characters.
- **CN**: This is the fully qualified domain name (FQDN) that specifies the server’s exact location in the Domain Name System (DNS)
- **subjectAltName**: From X509 Version 3 (RFC 2459), SSL certificates are allowed to specify multiple names that the certificate should match.
- If you plan to use the same certificate across multiple components, for example the Web Bridge, XMPP Server, Call Bridge and TURN server, then specify your domain name (DN) in the CN field, and in the SAN field specify your domain name (DN) and the FQDN for each of the components that will use the certificate.

Certificate Bundle

- A certificate bundle is a single file (with an extension of .pem, .cer or .crt) holding a copy of the Root CA's certificate and all intermediate certificates in the chain
- The certificates need to be in sequence with the certificate of the Root CA being last in the certificate bundle
- Combine all the certificates into one bundle of certificates in one of the following ways

- ❖ **Linux or Unix operating systems:**

```
cat cert1.cer cert2.cer cert3.cer > certbundle.cer
```

- ❖ **Windows or DOS:**

```
copy cert1.cer + cert2.cer + cert3.cer certbundle.cer
```

- ❖ You can create a certificate bundle by using a plain text editor such as notepad. All of the characters including the -----BEGIN CERTIFICATE- and -----END CERTIFICATE tags need to be inserted into the document. There should be no space between the certificates, for example no spaces or extra lines between -----END CERTIFICATE- of certificate 1 and -----BEGIN CERTIFICATE- of certificate 2. At the end of the file there should be 1 extra line
- ❖ Save the file with an extension of .pem, .cer, or .crt

CA vs Self signed Certificates

- Applications that interface internally within the Meeting Server only require certificates signed by an internal CA
- Internal CA signed certificates can be generated by a local or organizational Certificate Authority, such as an Active Directory server with the Active Directory Certificate Services Role installed
- The applications that require public CA signed certificates are
 - Webbridge: If using webRTC, WebRTC clients require a public CA signed certificate from the Web Bridge in order to trust the connection.
 - TURN server : If you configure TLS on your TURN server, then the TURN server will require a certificate/key pair similar to that created for the Web Bridge, so that the WebRTC client trusts the connection. The certificate should be signed by the same Certificate Authority as used for the Web Bridge certificate
 - Skype/Lync Integration: If you integrate CMS with Skype/Lync, then Callbridge certificate require to sign with same CA as Lync CA.

Lab 1

UC Infrastructure Integration

CUCM Integration

- CUCM Integration – Rendezvous (SIP trunk)
- CUCM Integration – Adhoc (conference bridge)

CUCM Integration - Rendezvous

- CUCM needs a trunk with the appropriate dial patterns or SIP route patterns in place to send the traffic to the CMS server.

SIP Trunk

☐		TO-CMS01-210	TO-CMS01-210	Default	CMS-RG	1	SIP Trunk	Full Service	Time In Full Service: 6 days 17 hours 4 minutes	Secure SIP Trunk Profile - CMS
☐		TO-CMS02-211	TO-CMS02-211	Default	CMS-RG	2	SIP Trunk	Full Service	Time In Full Service: 6 days 17 hours 4 minutes	Secure SIP Trunk Profile - CMS
☐		TO-CMS03-212	TO-CMS03-212	Default	CMS-RG	3	SIP Trunk	Full Service	Time In Full Service: 6 days 17 hours 4 minutes	Secure SIP Trunk Profile - CMS

Route Pattern

Route Patterns (1 - 5 of 5)										Rows
Find Route Patterns where Pattern begins with Find Clear Filter Add Remove										
☐	Pattern ^	Description	Partition	Route Filter	Associated Device					
☐	10200X				TO-GW-10					
☐	600XX				CUCM-TO-UCCECV					
☐	8000				CWMS-RL					
☐	99XX				10.124.56.210					

Note: SIP trunk need to reset after add on CUCM first time

CUCM Integration - Rendezvous (Cont'd)

- Ensure that the CMS inbound dial rules take into account the format the CUCM will be sending the URI in.

Incoming rule on CMS server

Incoming call handling

Call matching

☐	Domain name	Priority	Targets spaces	Targets IVRs	Targets Lync	
☐	10.124.56.210	100	yes	yes	no	no
☐	10.124.56.211	100	yes	yes	no	no
☐	10.124.56.212	100	yes	yes	no	no
☒	10.124.56.219	100	yes	yes	no	no
☐	10.124.56.220	100	yes	yes	no	no

CUCM Integration - Adhoc

- CMS can be added as a conference bridge in CUCM
- Once CMS registered as conference bridge, create appropriate MRG and MRGL
- Assign the MRGL to Phones

Conference Bridge Information
Conference Bridge : CFB-10.124.56.210
Registration: Registered with Cisco Unified Communications Manager cucsub01.cisco.lab
IPv4 Address: 10.124.56.210

Device Information
Conference Bridge Type* Cisco Meeting Server
☒ Device is trusted
Conference Bridge Name* CFB-10.124.56.210
Description
Conference Bridge Prefix
SIP Trunk* TO-CMS01-210
☐ Allow Conference Bridge Control of the Call Security Icon

HTTPS Interface Info
☐ Override SIP Trunk Destination as HTTPS Address
Hostname/IP Address
1 10.124.56.210
Username* admin
Password*
Confirm Password*
HTTPS Port* 8443

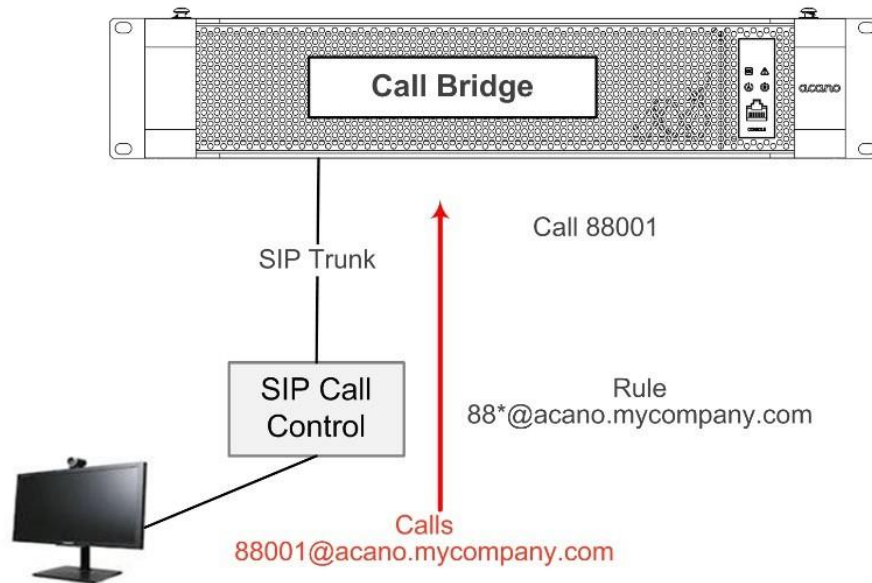
CUCM Integration - Certificates

- Rendezvous Calls
 - Upload Callbridge Certificate as CUCM-Trust
 - Upload RootCA/CA Bundle certificate as CUCM-Trust
- Adhoc Calls
 - Upload Webadmin certificate as CUCM-Trust
 - Upload RootCA/CA Bundle certificate as CUCM-Trust

Note: Wild card certificates are not supported. Certificates must have SSL Client and SSL Server Purposes enabled

VCS/Expressway Integration

- Implemented via a neighbor zone pointing to the CMS call-bridge.
- Outbound domain can be transformed via search rule if necessary for inbound matching.
- VCS should also be configured to handle incoming calls appropriately.



VCS/Expressway Integration (Cont'd)

Search Rule

Configuration

Rule name	* Acano
Description	To Acano Bridge
Priority	* 43
Protocol	Any
Source	Any
Request must be authenticated	No
Mode	Alias pattern match
Pattern type	Regex
Pattern string	*.*@acanolab.tkratzke.local
Pattern behavior	Leave
On successful match	Continue
Target	* Acano
State	Enabled

Save Delete Cancel

Neighbor Zone

Configuration

Name	* Acano
Type	Neighbor
Hop count	* 15

H.323

Mode	Off
------	-----

SIP

Mode	On
Port	* 5060
Transport	TCP
Accept proxied registrations	Allow
Media encryption mode	Auto
ICE support	Off

Authentication

Authentication policy	Treat as authenticated
SIP authentication trust mode	Off

Location

Peer 1 address	14.80.99.225	SIP: Reachable: 14.80.99.225:5060
Peer 2 address		

Lync (Skype for Business) Integration call types

Gateway : call goes across CMS for adaptation and transcoding



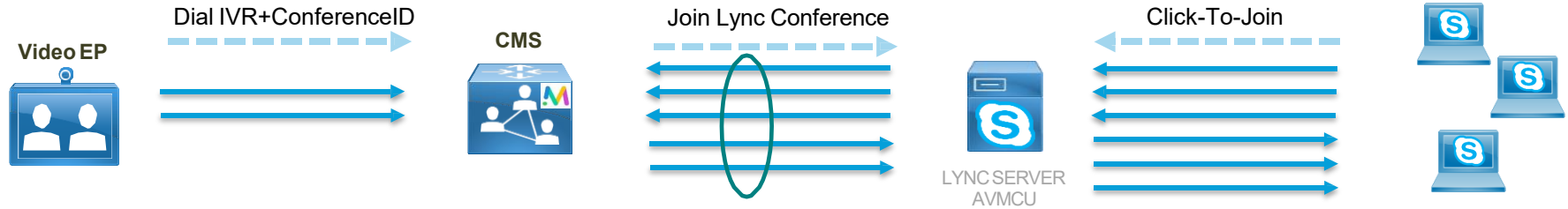
Rendezvous : Video and Lync Endpoints meet in a CMS space



Dual Homed conferencing – Video endpoint joins via CMS and Lync/S4B meeting as a full featured participants (picture in the next slides)

Lync Integration (Dual Homed)

-Dial Lync meeting via IVR



Dial Plan Overview

Local Dial Plan

- **Outbound Call** webadmin Configuration describes:
 - Where CMS server send a call based on domain in URI
 - Any transformation that need to be done on URI
- **Incoming Call** webadmin configuration describes if incoming call need to be :
 - Handled locally (spaces, users)
 - Forwarded to an external destination (e.g. call from CUCM to Lync) via dial plan (Outbound Call) configuration

(forward behaviour can be summarized as below:

***Incoming Call** => **Incoming Call** rule (no match) => **Forwarded Call** rule (match) => handle call according to **Outbound Call** rule)*

Local Dial Plan - Outbound (Cont'd)

Domain: Destination Domain of the outgoing call

SIP Proxy: which SIP trunk to use

Local Contact Domain: to be used only with Lync, FQDN of CMS server
(applies to the domain of the “Contact” header in the outgoing SIP INVITE, if blank IP address of the server is used)

Local From Domain: domain used as “From”
(applies to the domain of the “From” header of the outgoing SIP INVITE, if blank domain of the server is used)

Trunk Type: Standard SIP or Lync or Avaya

Encryption Type: Auto or Encrypted or Unencrypted (use Encrypted for Lync trunks)

Priority: order of priority of the rule (high priority is tried first)

Outbound calls

Filter 提交查询											
☐	Domain	SIP proxy to use	Local contact domain	Local from domain	Trunk type	Behavior	Priority	Encryption	Tenant	Call Bridge Scope	
☐	<match all domains>	<none; call directly>		<use local contact domain>	Standard SIP	Stop	0	Auto	no	<all>	[edit]
☒	10.124.56.210	<none; call directly>		<use local contact domain>	Standard SIP	Stop	0	Auto	no	<all>	[edit]
☐	10.124.56.211	<none; call directly>		<use local contact domain>	Standard SIP	Stop	0	Auto	no	<all>	[edit]
☐	10.124.56.212	<none; call directly>		<use local contact domain>	Standard SIP	Stop	0	Auto	no	<all>	[edit]
☐	10.124.56.219	<none; call directly>		<use local contact domain>	Standard SIP	Stop	0	Auto	no	<all>	[edit]
☐	10.124.56.220	<none; call directly>		<use local contact domain>	Standard SIP	Stop	0	Auto	no	<all>	[edit]
☐	cisco.lab	cucmpub.cisco.lab		<use local contact domain>	Standard SIP	Stop	0	Auto	no	<all>	[edit]
☐	cisco.lab	cucmsub.cisco.lab		<use local contact domain>	Standard SIP	Stop	0	Auto	no	<all>	[edit]
					Standard SIP v	Stop v	0	Auto v			Add New Reset

1

[Delete](#)

Local Dial Plan - Inbound (Cont'd)

When an incoming rule is defined

Domain: with domain the callbridge consider “own” and use to match users and spaces (similar to CUCM “Cluster Fully Qualified Domain Name”)

Priority has the usual meaning

Target: **Spaces**, **User**, **IVR** or **Lync** can be used to specify better the scope

Which domain name should be used for incoming calls?

1. Domain assigned to callbridge server
2. IP address/FQDN of all call bridges in cluster
(*CUCM uses it for call routed with numeric patterns*)
3. Lync Domain (-> *to be used with target Lync*)

Incoming call handling

Call matching

☐	Domain name	Priority	Targets spaces	Targets IVRs	Targets Lync	Targets Lync Simplejoin	Tenant	
☐	10.124.56.210	100	yes	yes	no	no	no	[edit]
☐	10.124.56.211	100	yes	yes	no	no	no	[edit]
☐	10.124.56.212	100	yes	yes	no	no	no	[edit]
☐	10.124.56.219	100	yes	yes	no	no	no	[edit]
☐	10.124.56.220	100	yes	yes	no	no	no	[edit]
☐	cobn.cn	100	yes	yes	no	no	no	[edit]
☐	cisco.lab	100	yes	yes	no	no	no	[edit]
		0	yes	yes	no	no		Add New Reset

1

Call forwarding

☐	Domain matching pattern	Priority	Forward	Caller ID	Rewrite domain	Forwarding domain	
		0	reject	use dial plan	no		Add New Reset

Additional Features

CDRs

- CDRs cannot be viewed directly on the CMS server.
- Sent via HTTP or HTTPS as a series of XML documents to the configured server.
- CDR destination is configured under the configuration -> CDR section of the webadmin.
- If CMM is in use it must be configured as the CDR destination.

CDR configuration

CDR receiver settings

Receiver URI 1

https://10.124.56.209/meetings/cdr/1/?token=601a12

Receiver URI 2

Receiver URI 3

Receiver URI 4

Submit

License Model

- Required Activation Keys and Licenses
- Activation Keys are needed for
 - Call-Bridge
 - Recording
 - Streaming
- Licenses
 - Multiparty Licensing (PMP Plus and SMP Plus)
 - Cisco Capacity Units
- Licenses are tied to MAC Address

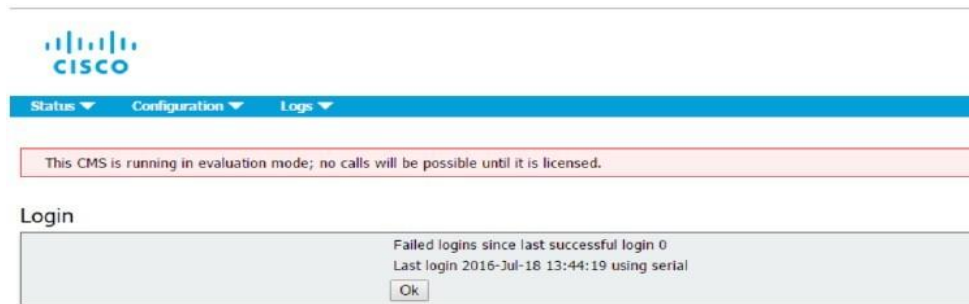


Table 1: Capacity Unit Licensing

Media Stream	Number of licenses per Capacity Unit	Number of licenses required per call leg
1080p30	0.5	2
720p30	1	1
480p30	2	0.5

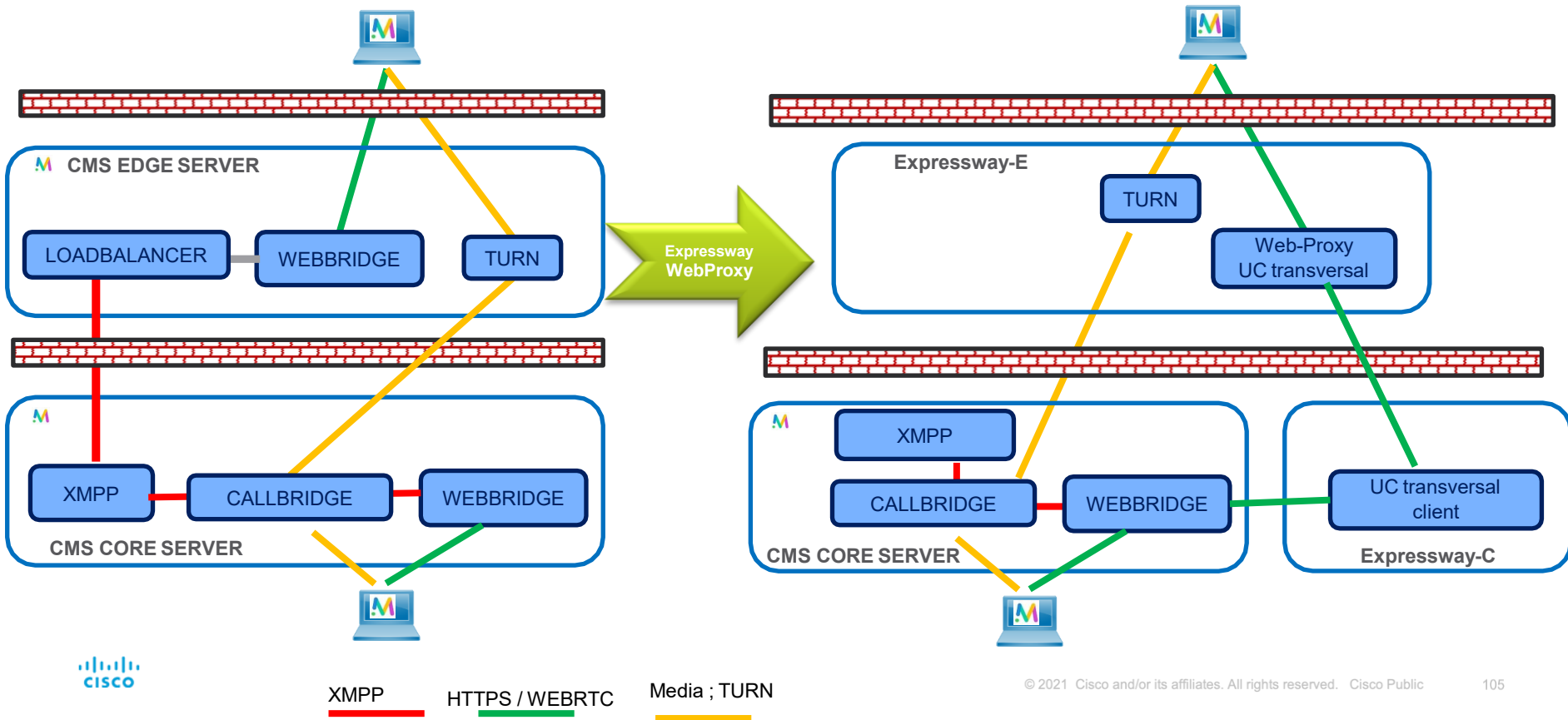
Expressway proxy for WebRTC

Cisco Meeting Server Web Proxy

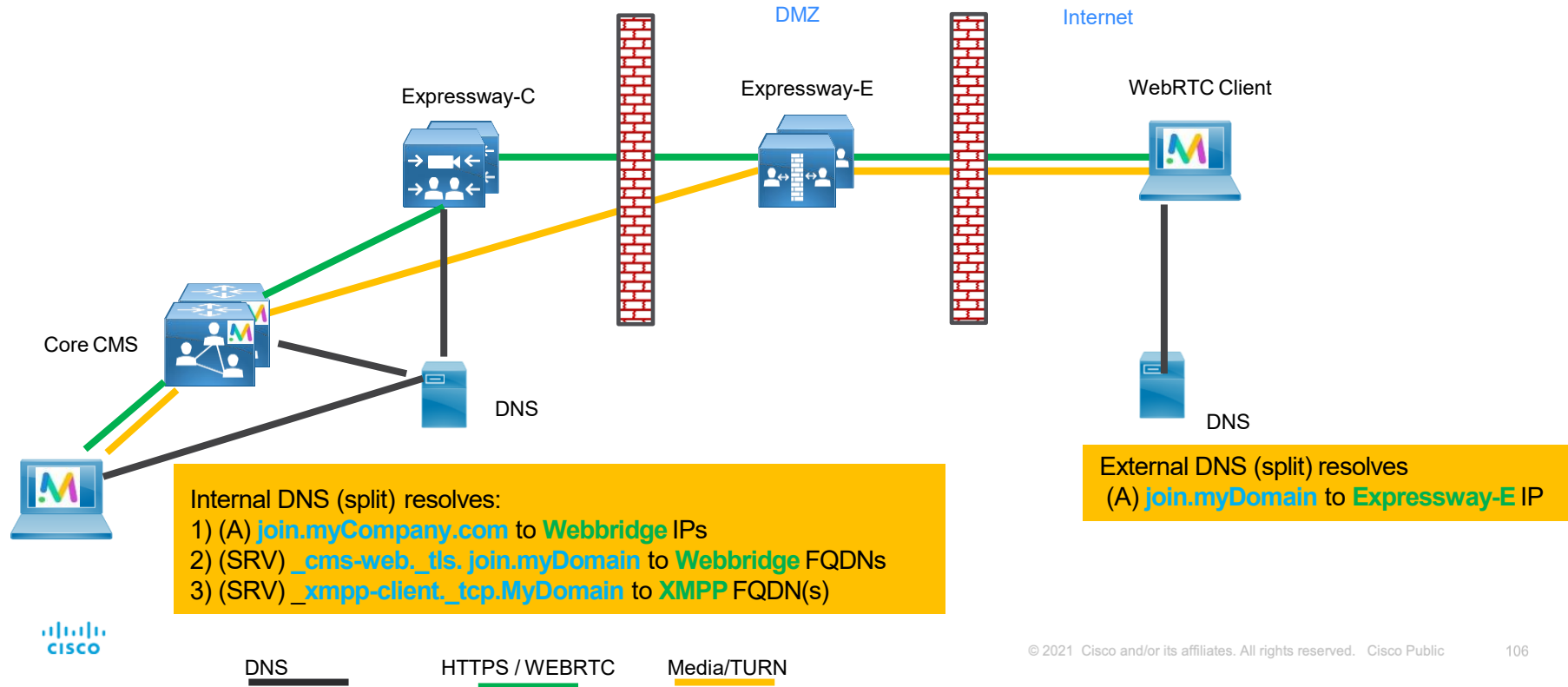
Cisco Meeting Server Web Proxy is supported From Expressway X8.9.2 release

- CMS WebProxy feature in Expressway allows to leverage expressway existing capabilities (TURN and MRA Web-Proxy) to facilitate firewall transversal of WebRTC CMS clients
- On expressway it can coexist with MRA, B2B, Registrar, IMP federation but not with Jabber-Guest or MS Interop
- CMA native clients still need classical CMS edge
- Expressway TURN supports TCP 443 as TURN port from x8.11 onwards

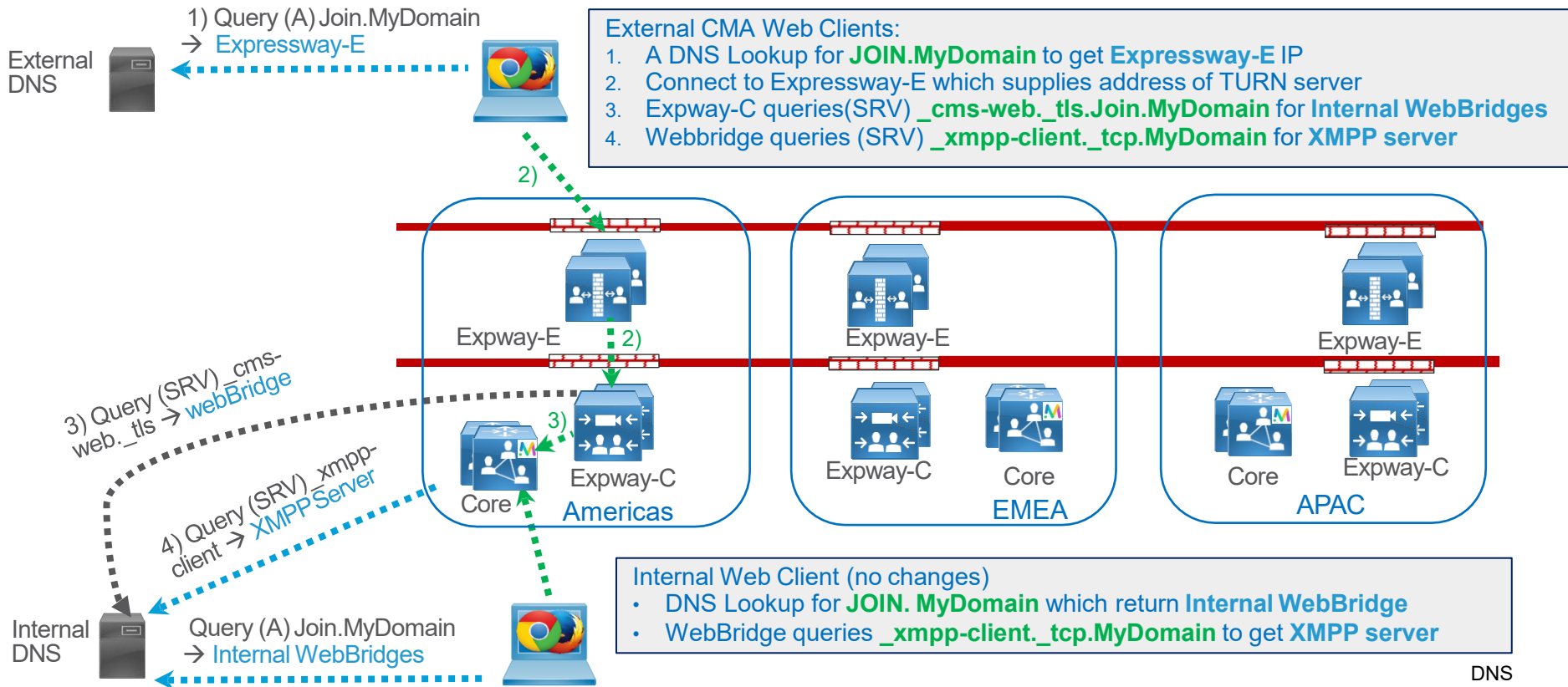
From legacy CMS Edge to Expressway Web-Proxy



Cisco Meeting Server Web Proxy

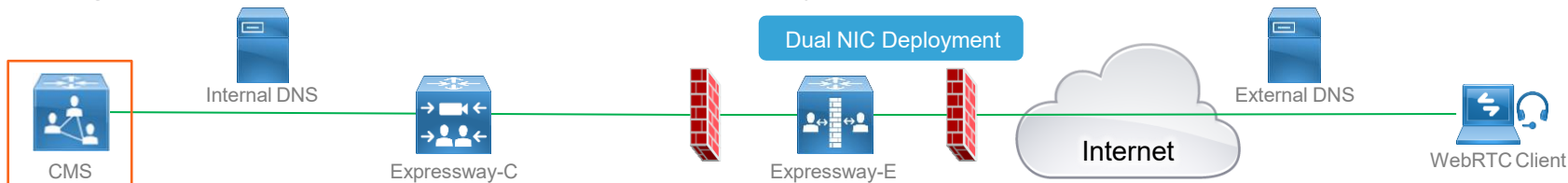


DNS design - Web Client with Expressway



Cisco Meeting Server Web Proxy

- CMS Configuration - TURN Server
 - Configure TURN server on CMS point to Expressway-E

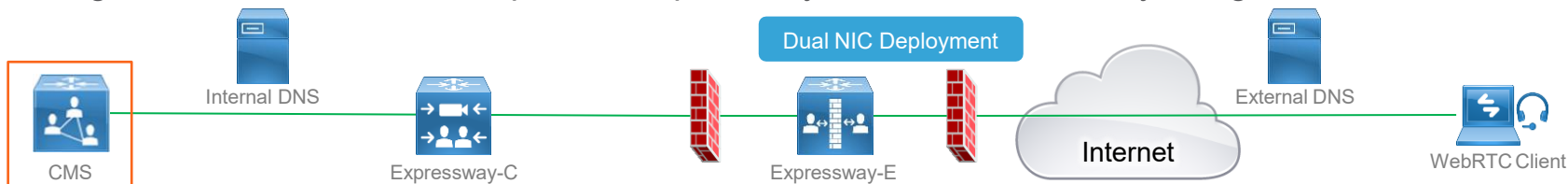


The screenshot shows the Cisco Meeting Server Web GUI. The top navigation bar includes 'Status', 'Configuration', and 'Logs'. The 'Configuration' tab is selected. The 'General configuration' section is visible, and the 'TURN Server settings' section is highlighted with a large red X. The settings include fields for 'TURN Server address (CMS)', 'TURN Server address (CMA)', 'Username', 'Password', and 'Confirm password'.

DO NOT configure TURN parameters from Web GUI

Cisco Meeting Server Web Proxy

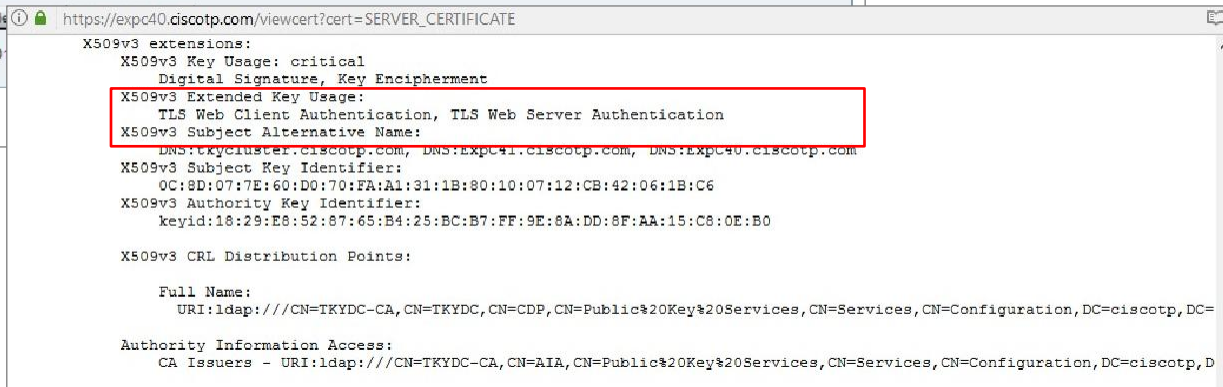
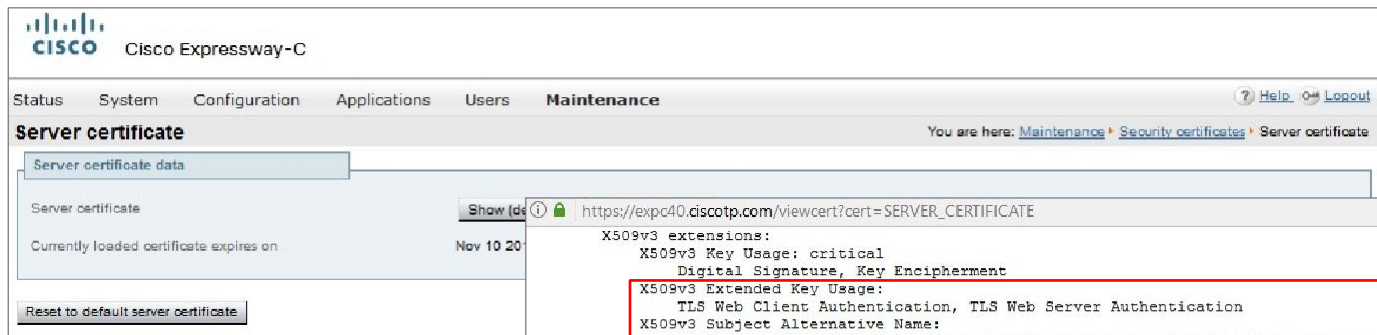
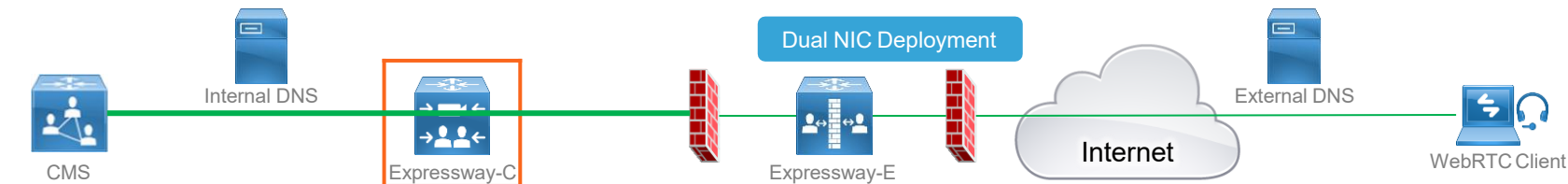
- CMS Configuration - TURN Server
 - Configure TURN server on CMS point to Expressway-E as TURN server by using API



Parameter	Value
serverAddress	TURN Server's FQDN/IP Address (Expressway-E Private IP Address, i.e. address of LAN1)
clientAddress	TURN Server's FQDN/IP Address (Expressway-E Public IP Address, i.e. NAT address of LAN2)
username	TURN Authentication realm
password	TURN Authentication password
type	expressway
tcpPortNumberOverride	443

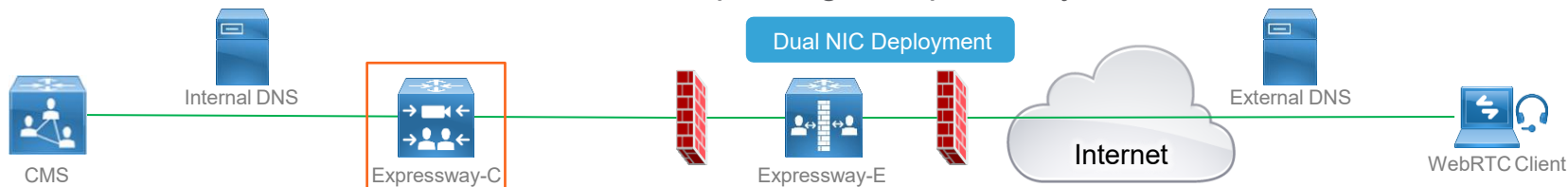
Cisco Meeting Server Web Proxy

- Expressway-C Configuration - Certification
- Install trusted CA certificate and server certificate



Cisco Meeting Server Web Proxy

- Expressway-C Configuration - UC Traversal Zone
- Create Unified Communications traversal zone pointing to Expressway-E



Cisco Expressway-C

Status System **Configuration** Applications Users Maintenance [Help](#) [Logout](#)

Edit zone You are here: [Configuration](#) > [Zones](#) > [Zones](#) > [Edit zone](#)

Search rules not configured: This zone does not appear as the target in any zone search rules. You can configure which zones are searched and in what order on the [Search rules](#) page.

Configuration

Name * TKYExpE26-UC ⓘ

Type Unified Communications traversal

Hop count * 15 ⓘ

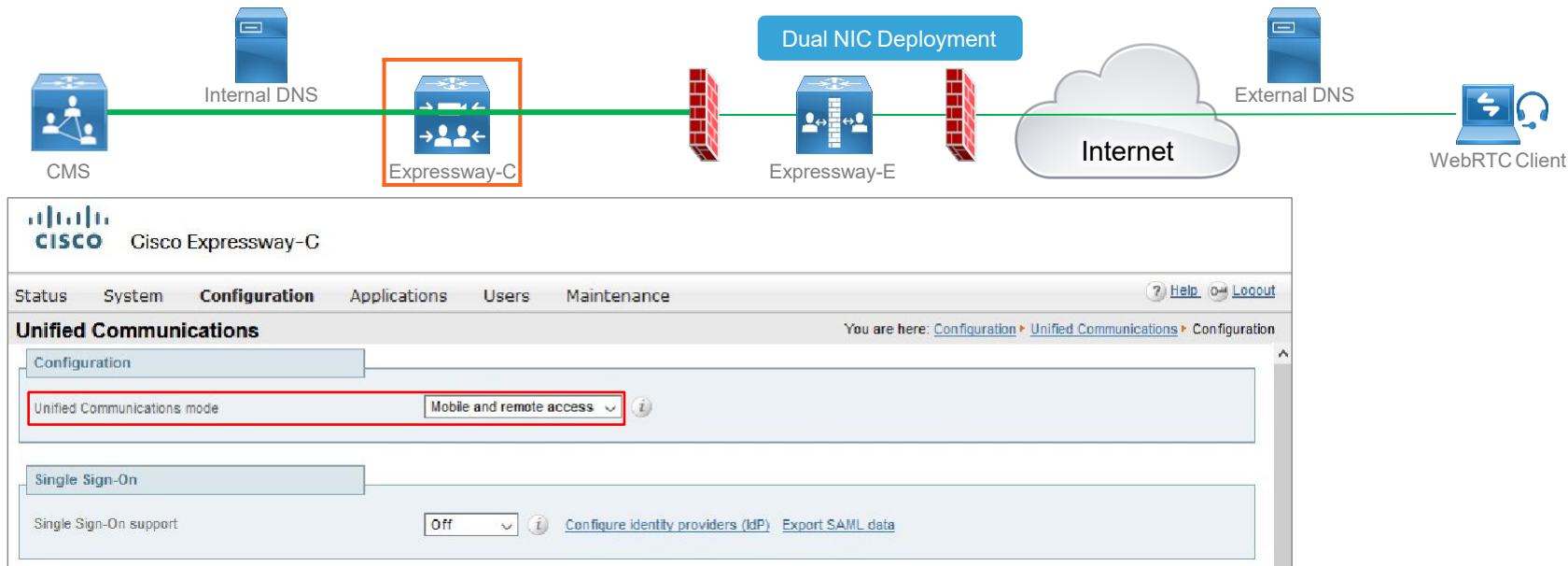
Connection credentials

Username * UCTGExpUC ⓘ

Password * ⓘ

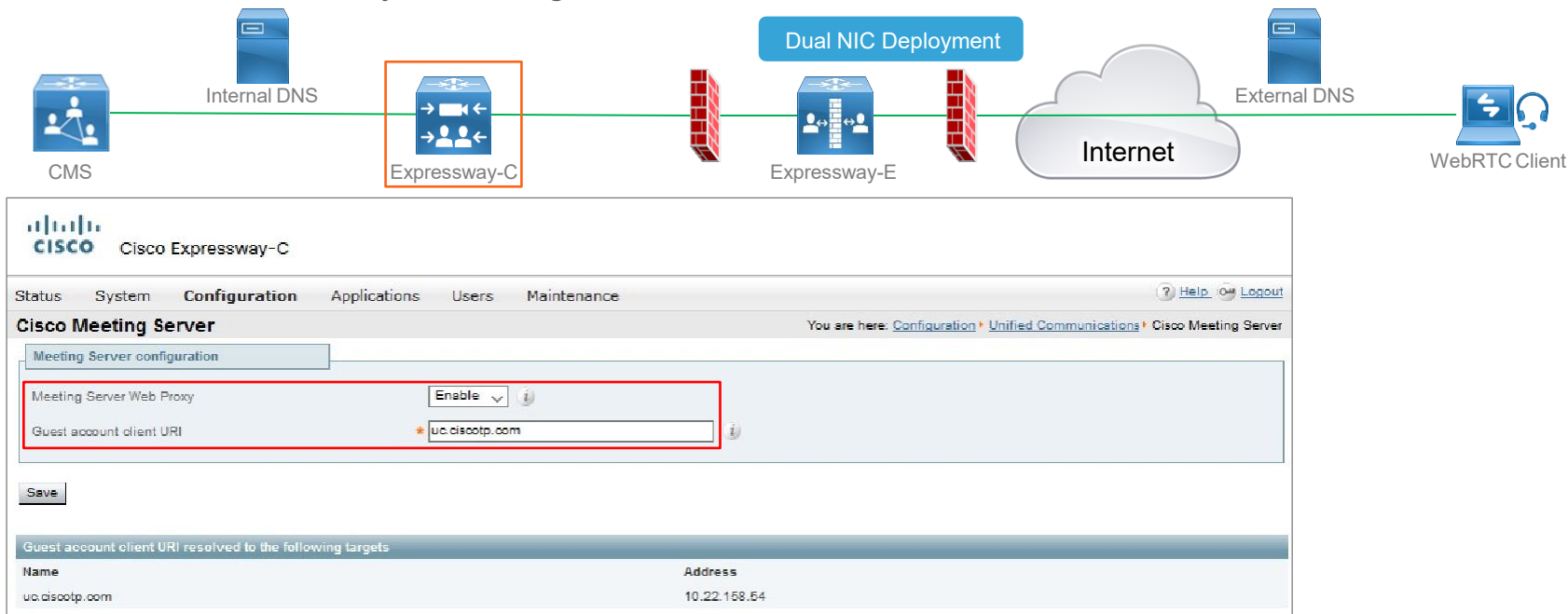
Cisco Meeting Server Web Proxy

- Expressway-C Configuration - Unified Communications
 - Enabled Mobile and Remote Access mode



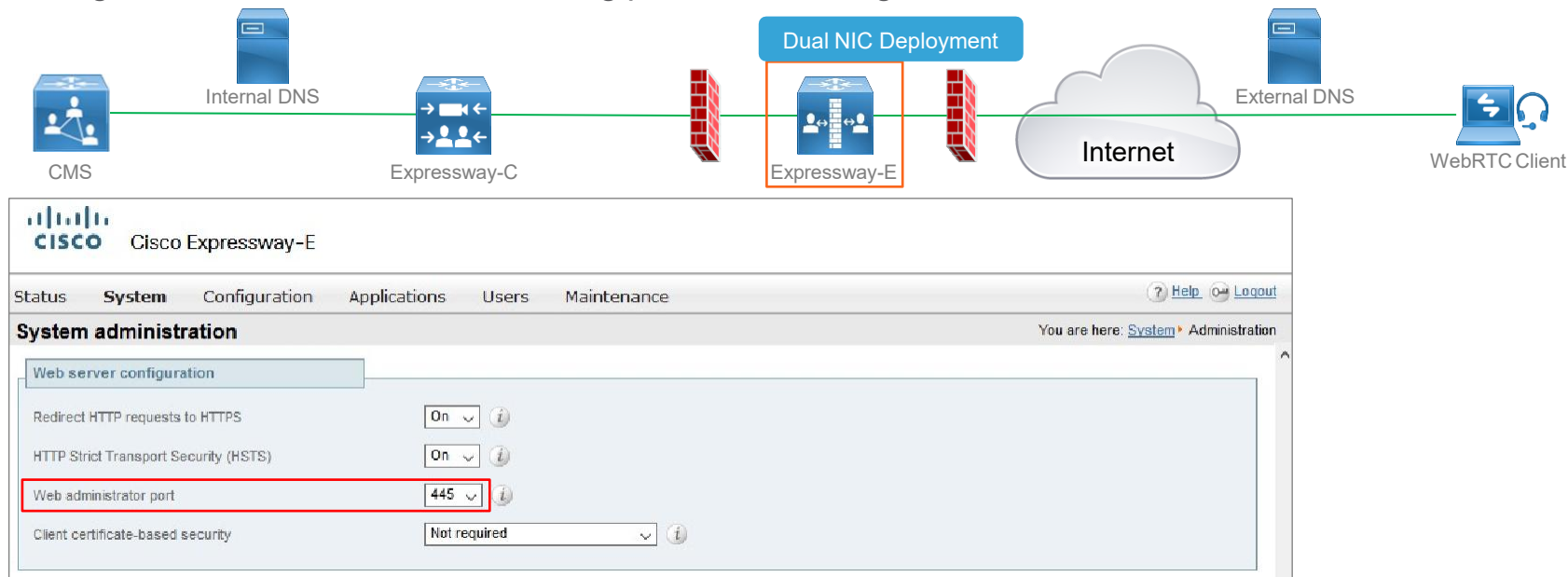
Cisco Meeting Server Web Proxy

- Expressway-C Configuration - Guest account client URI
 - Enabled CMS Web Proxy and configure Guest account client URI



Cisco Meeting Server Web Proxy

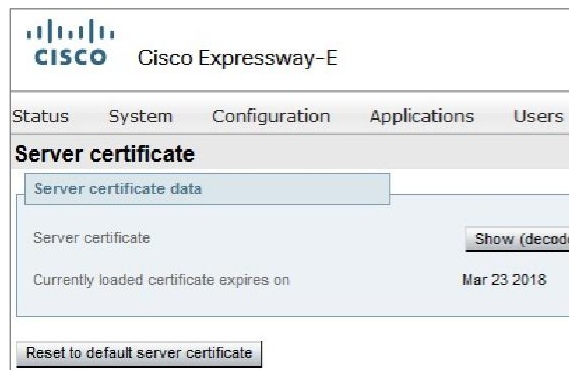
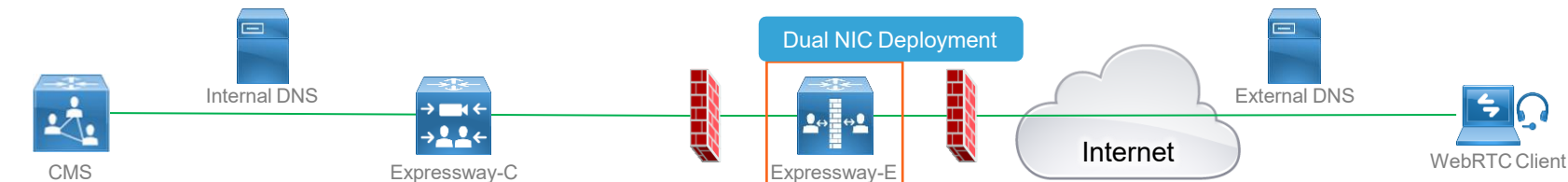
- Expressway-E Configuration - Web Administration Listening Port
 - Change the web administration listening port to something other than default 443



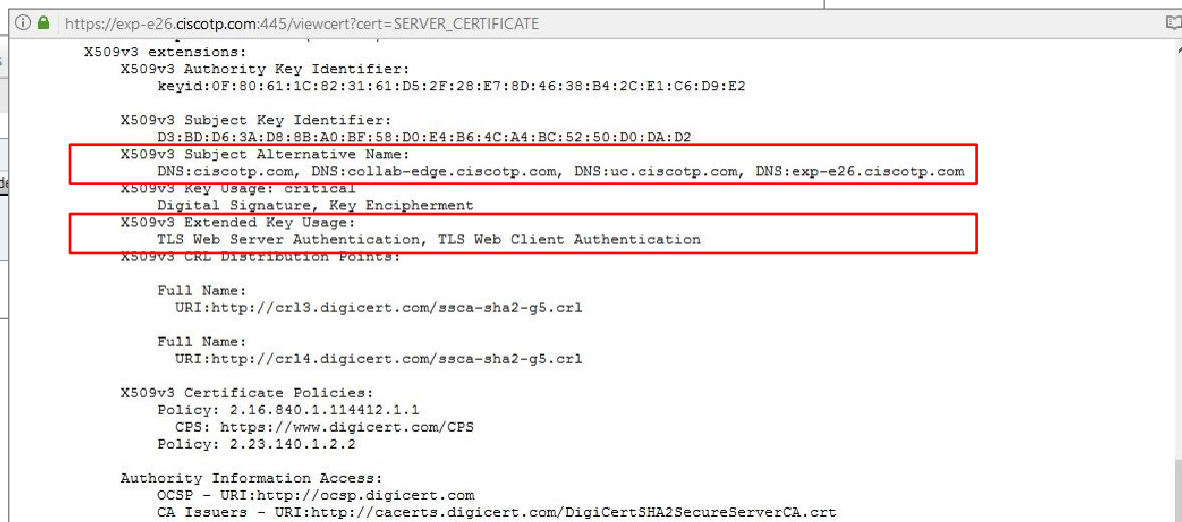
➤ In default, Web GUI only allow to change 443 or 445, but support to modify other port by using CLI

Cisco Meeting Server Web Proxy

- Expressway-E Configuration - Certification
- Install trusted CA certificate and server certificate

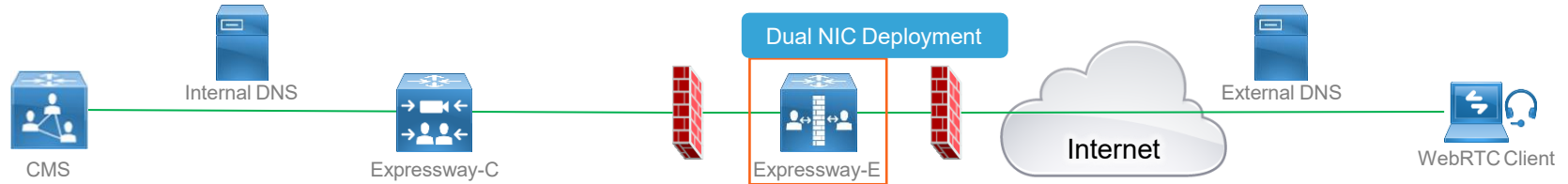


- Guest account client URI should be added on Expressway-E certificate as additional SAN

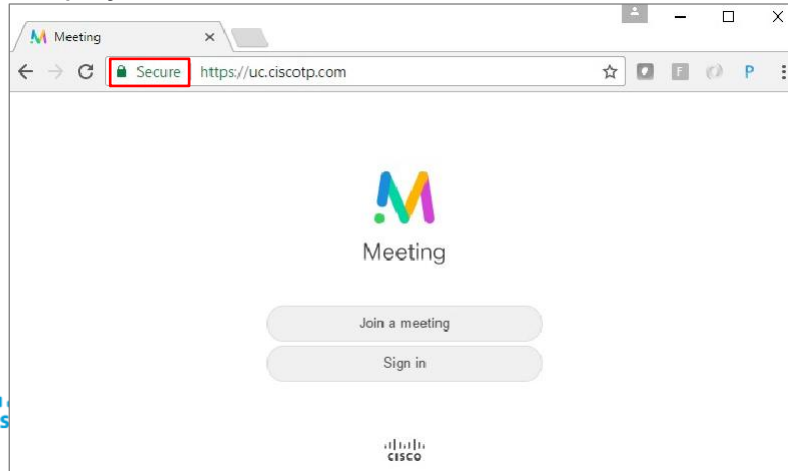


Cisco Meeting Server Web Proxy

- Expressway-E Configuration - Certification
 - Install trusted CA certificate and server certificate

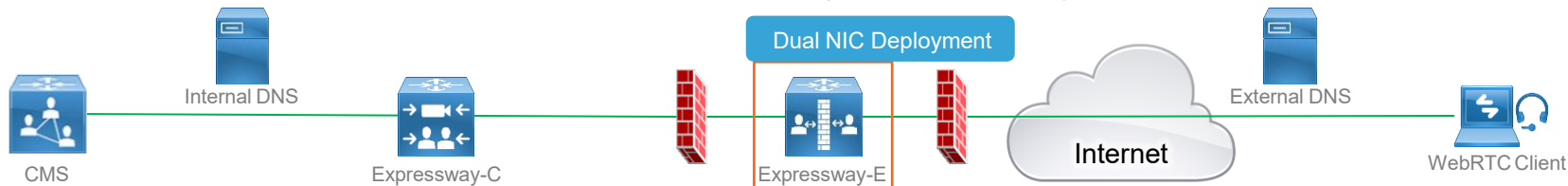


- Guest account client URI should be added on Expressway-E certificate as additional SAN for WebRTC session to display as secure connection



Cisco Meeting Server Web Proxy

- Expressway-E Configuration - UC Traversal Zone
- Create Unified Communications traversal zone pointing to Expressway-E



Cisco Expressway-E

Status System **Configuration** Applications Users Maintenance [Help](#) [Logout](#)

Edit zone You are here: [Configuration](#) > [Zones](#) > [Zones](#) > Edit zone

Search rules not configured: This zone does not appear as the target in any zone search rules. You can configure which zones are searched and in what order on the [Search rules](#) page.

Configuration

Name ⓘ

Type Unified Communications traversal

Hop count ⓘ

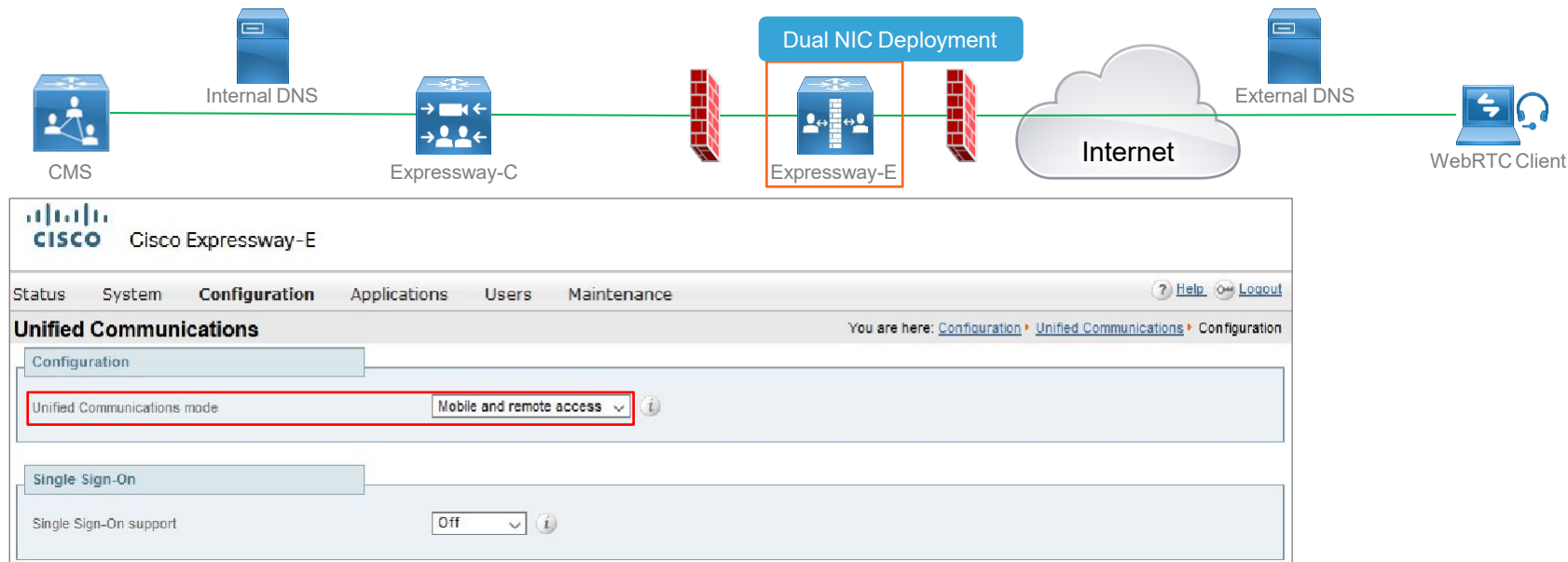
Connection credentials

Username ⓘ

Password [Add/Edit local authentication database](#)

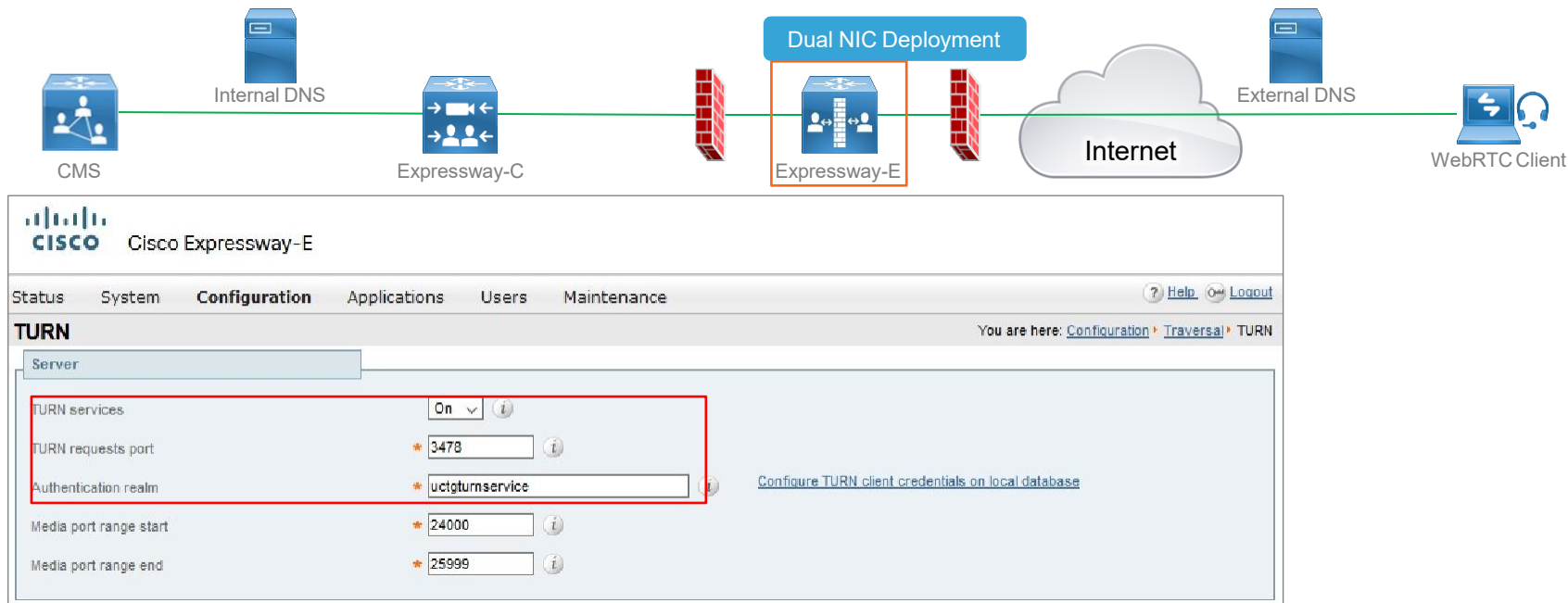
Cisco Meeting Server Web Proxy

- Expressway-E Configuration - Unified Communications
 - Enabled Mobile and Remote Access mode



Cisco Meeting Server Web Proxy

- Expressway-E Configuration - Unified Communications
 - Enabled Mobile and Remote Access mode



Logs and Troubleshooting Tools

CLI (MMP) Basic Troubleshoot Commands

```
Core> ping ?  
Elicit ICMP ECHO_RESPONSES for a remote host
```

```
Usage:  
ping <target>  
Core>  
Core> ping 14.80.99.228  
PING 14.80.99.228 (14.80.99.228): 56 data bytes  
64 bytes from 14.80.99.228: seq=0 ttl=64 time=0.253 ms  
64 bytes from 14.80.99.228: seq=1 ttl=64 time=0.347 ms  
64 bytes from 14.80.99.228: seq=2 ttl=64 time=0.192 ms  
64 bytes from 14.80.99.228: seq=3 ttl=64 time=0.255 ms  
--- 14.80.99.228 ping statistics ---  
4 packets transmitted, 4 packets received, 0% packet loss  
round-trip min/avg/max = 0.192/0.261/0.347 ms
```

```
Core> traceroute ?  
Track route of packets to a remote host
```

```
Usage:  
traceroute <target>  
Core>  
Core> traceroute 14.80.79.2  
traceroute to 14.80.79.2 (14.80.79.2), 30 hops max, 46 byte packets  
 1 14.80.99.1 (14.80.99.1) 0.421 ms 0.474 ms 0.405 ms  
 2 14.80.79.2 (14.80.79.2) 1.521 ms 0.452 ms 0.432 ms  
Core> █
```

```
cmslab1> ipv4 a  
IPv4 configuration:  
address 10.104.215.252  
default true  
dhcp false  
enabled true  
gateway 10.104.215.193  
macaddress 00:50:56:AD:25:9C  
prefixlen 26  
IPv4 observed values  
Addresses:  
10.104.215.252/26  
Routes:  
source destination gateway global  
0.0.0.0 10.104.215.252 0.0.0.0 false  
0.0.0.0 10.104.215.192 0.0.0.0 false  
0.0.0.0 10.104.215.192 0.0.0.0 false  
0.0.0.0 10.104.215.255 0.0.0.0 false  
0.0.0.0 10.104.215.193 10.104.215.193 true  
cmslab1>  
cmslab1> █
```

- Ping
 - “ping <destination>”
- Traceroute
 - “traceroute <destination>”
- DNS Lookup
 - “dns lookup <A/AAAA/SRV>
<hostname>”

```
cmslab1> ipv4 ?  
Configure IPv4 addresses
```

```
Usage:  
ipv4 <interface>  
ipv4 <interface> dhcp  
ipv4 <interface> disable  
ipv4 <interface> enable  
ipv4 <interface> add <address>/<prefix length> <gateway>  
ipv4 <interface> del <address>  
ipv4 <interface> default  
ipv4 <interface> route add <address>/<prefix length>  
ipv4 <interface> route del <address>/<prefix length>
```

Troubleshooting Tools

- WebAdmin Logs
- SFTP Files
- SFTP: `logbundle.tar.gz` (Version 2.2+)
- pcaps

WebAdmin Logs

- Two places to view errors/logs
 - Webadmin “status” page
 - Webadmin “logs” page
- The status page only shows specifics errors or warnings
- Webadmin page viewed logs do not contain all logging info



Status	Configuration	Logs
System status		
Uptime	26 days, 16 hours, 23 minutes	
Build version	2.2.7	
XMPP connection	connected to 10.104.215.211 (secure) for 26 days, 15 hours, 3 minutes	
Authentication service	registered for 26 days, 15 hours, 3 minutes	
Lync Edge registrations	not configured	
CMA calls	0	
SIP calls	0	
Lync calls	0	
Forwarded calls	0	
Completed calls	0	
Activated conferences	0	
Active Lync subscribers	0	
Total outgoing media bandwidth	0	
Total incoming media bandwidth	0	

Fault conditions

Date	Time	Fault condition
2018-01-20	23:17:23.802	Web bridge connection to "meet.cmsclusterlab.com" failed (DNS failure)

Recent errors and warnings

Date	Time	Logging level	Message
2018-01-20	23:17:26.803	Warning	web bridge link 2: name resolution for "meet.cmsclusterlab.com" failed
2018-01-20	23:17:23.802	Warning	web bridge link 2: name resolution for "meet.cmsclusterlab.com" failed
2018-01-19	14:02:33.355	Warning	remote link to "core2": no progress after 60008ms
2018-01-19	11:05:01.673	Warning	invalid API operation - unhandleable URI
2018-01-19	11:04:54.852	Warning	invalid API operation - unhandleable URI
2018-01-17	20:01:03.847	Warning	invalid API operation - unhandleable URI



Status▼Configuration▼

Filter

Logs▲

Event log

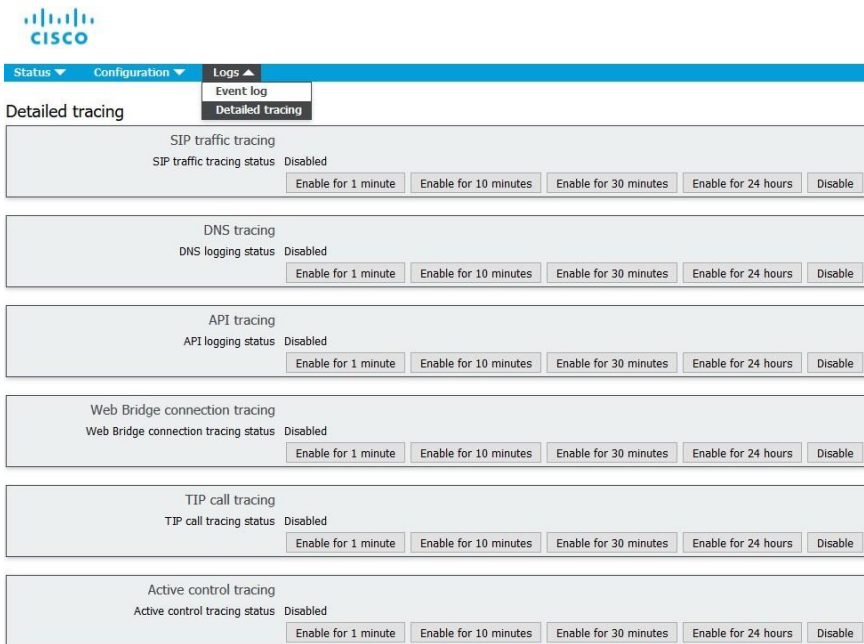
Detailed tracing

Date	Time	Logging level	
2017-04-13	20:21:38.366	Info	53 log messages cleared by "admin"
2017-04-13	20:22:22.465	Info	web bridge link 1 connection established
2017-04-13	20:22:22.465	Info	web bridge link 1: configuration request received (custom background url
2017-04-13	20:22:23.465	Info	web bridge link 1: configuration request received (custom background url
2017-04-13	20:22:23.488	Info	failed to establish connection to web bridge link 1 (service unavailable)
2017-04-13	20:22:35.465	Info	web bridge link 1 connection established
2017-04-13	20:33:22.631	Info	web user "admin" log in timed out
2017-04-13	20:33:22.631	Info	web session 1 for user "admin": deactivating due to time out
2017-04-13	20:33:22.631	Info	web session 1 deactivated for user "admin"
2017-04-13	20:35:51.767	Info	media module status 1



Logging Levels

- In WebAdmin we can increase the logging level of certain services
- If logging is not enabled, detailed information will not be written to the syslog
- Logging automatically disables after the timer expires



The screenshot displays the Cisco WebAdmin interface for configuring logging levels. The top navigation bar includes 'Status', 'Configuration', and 'Logs'. Under 'Logs', 'Event log' and 'Detailed tracing' are visible, with 'Detailed tracing' selected. The 'Detailed tracing' section contains several configuration panels, each with a service name, a status indicator, and a set of buttons to enable logging for different durations or to disable it.

Service
SIP traffic tracing
SIP traffic tracing status
Disabled
Enable for 1 minute
Enable for 10 minutes
Enable for 30 minutes
Enable for 24 hours
Disable

Service
DNS tracing
DNS logging status
Disabled
Enable for 1 minute
Enable for 10 minutes
Enable for 30 minutes
Enable for 24 hours
Disable

Service
API tracing
API logging status
Disabled
Enable for 1 minute
Enable for 10 minutes
Enable for 30 minutes
Enable for 24 hours
Disable

Service
Web Bridge connection tracing
Web Bridge connection tracing status
Disabled
Enable for 1 minute
Enable for 10 minutes
Enable for 30 minutes
Enable for 24 hours
Disable

Service
TIP call tracing
TIP call tracing status
Disabled
Enable for 1 minute
Enable for 10 minutes
Enable for 30 minutes
Enable for 24 hours
Disable

Service
Active control tracing
Active control tracing status
Disabled
Enable for 1 minute
Enable for 10 minutes
Enable for 30 minutes
Enable for 24 hours
Disable

SFTP Files

- Certificates and Keys
- live.json
- Boot.json
- Backups
- Audit log
- Log (syslog)
- Crash dumps
- pcaps

Name	Size	Changed	Rights	Owner
 ACANO-MIB.txt	4 KB	3/29/2017 5:05:09 PM	r--r--r--	admin
 ACANO-SYSLOG-MIB.txt	2 KB	3/29/2017 5:23:32 PM	r--r--r--	admin
 admin-a-20170404-202853.pcap	1 KB	4/4/2017 4:28:54 PM	r--r--r--	admin
 audit	33,757 KB	4/17/2017 12:18:38 PM	r--r--r--	admin
 boot.json	38 KB	4/3/2017 10:44:20 AM	r--r--r--	admin
 callbridge.crt	39 KB	3/17/2016 3:44:06 PM	r--r--r--	admin
 callbridge.key	39 KB	3/17/2016 3:44:06 PM	r--r--r--	admin
 cms.lic	39 KB	4/17/2017 12:21:14 PM	r--r--r--	admin
 live.json	39 KB	4/17/2017 12:21:14 PM	r--r--r--	admin
 log	67,156 KB	4/17/2017 12:20:59 PM	r--r--r--	admin
 logbundle.tar.gz	1 KB	4/17/2017 12:21:15 PM	r--r--r--	admin
 upgrade.img	213,212 KB	4/3/2017 10:43:40 AM	r--r--r--	admin
 web.crt	39 KB	3/10/2016 11:46:20 AM	r--r--r--	admin
 web.key	39 KB	3/10/2016 11:46:20 AM	r--r--r--	admin
 webbridge.crt	39 KB	3/22/2016 6:39:26 PM	r--r--r--	admin
 webbridge.key	39 KB	3/22/2016 6:39:26 PM	r--r--r--	admin
 xmpp.crt	39 KB	3/22/2016 6:31:58 PM	r--r--r--	admin
 xmpp.key	39 KB	3/22/2016 6:31:58 PM	r--r--r--	admin

Log Bundle

- Version 2.2 of CMS adds a new log bundle that can be collected via SFTP
- Syslog
- Database
- Crash Dumps
- Live.json

Name	Size	Changed	Rights	Owner
 ACANO-MIB.txt	4 KB	3/29/2017 5:05:09 PM	r--r--r--	admin
 ACANO-SYSLOG-MIB.txt	2 KB	3/29/2017 5:23:32 PM	r--r--r--	admin
 audit	12,568 KB	4/17/2017 12:07:05 PM	r--r--r--	admin
 boot.json	103 KB	4/13/2017 3:03:11 PM	r--r--r--	admin
 cms.lic	108 KB	4/17/2017 12:07:01 PM	r--r--r--	admin
 cmsrtp1.cer	108 KB	4/13/2017 4:15:13 PM	r--r--r--	admin
 cmsrtp1.csr	108 KB	4/13/2017 4:07:00 PM	r--r--r--	admin
 cmsrtp1.key	108 KB	4/13/2017 4:07:00 PM	r--r--r--	admin
 combinedserver.cer	108 KB	4/13/2017 4:31:07 PM	r--r--r--	admin
 live.json	108 KB	4/17/2017 12:07:01 PM	r--r--r--	admin
 log	94,024 KB	4/17/2017 12:07:05 PM	r--r--r--	admin
 logbundle.tar.gz	1 KB	4/17/2017 12:07:05 PM	r--r--r--	admin
 upgrade.img	213,212 KB	4/4/2017 12:10:03 PM	r--r--r--	admin

 db		File folder	
 dumps		File folder	
 .cms	12	CMS File	12/31/1969 7:0...
 live.json	30,683	JSON File	12/31/1969 7:0...
 syslog	16,333,022	File	4/17/2017 12:0...
 syslog.1	52,429,041	1 File	12/4/2016 11:3...

pcap–Packet Capture

- Pcap <interface>
- Show all network traffic in/out an interface
- Extremely useful in troubleshooting network impairment
- Captures roughly 1GB before ending
- Capture command issued via MMP
- File downloaded via SFTP

Protocol	Length	Info
RTP	214	PT=DynamicRTP-Type-114, SSRC=0xB2B0B0C5, Seq=415, Time=966169251
RTP	214	PT=DynamicRTP-Type-114, SSRC=0xB2B0B0C5, Seq=416, Time=966170211
RTP	214	PT=DynamicRTP-Type-114, SSRC=0xB2B0B0C5, Seq=417, Time=966171171
RTP	214	PT=DynamicRTP-Type-114, SSRC=0xB2B0B0C5, Seq=418, Time=966172131
RTP	214	PT=DynamicRTP-Type-114, SSRC=0xB2B0B0C5, Seq=419, Time=966173091
RTP	214	PT=DynamicRTP-Type-114, SSRC=0xB2B0B0C5, Seq=420, Time=966174051

CLI (MMP) Backup/Restore Files

- Full system snapshot – Creates a full snapshot of entire config along with database for download. This includes spaces, chat, and activity.

“backup snapshot <filename>”

- Restore full snapshot...

“backup rollback <filename>”

- *“Reboot”*

- List all current backups on system

“backup list”

```
Core> backup snapshot fullbackup
fullbackup.bak ready for download
Core>
Core>
Core> backup list
Apr 13 16:13 backup_Wed_Apr_13_16_10_26_2016
Apr 13 16:23 backup_Wed_Apr_13_16_20_24_2016
Apr 13 16:38 backup_Wed_Apr_13_16_34_49_2016
Apr 13 16:47 backup_Wed_Apr_13_16_43_29_2016
Apr 19 13:46 backup_Tue_Apr_19_13_41_51_2016
Apr 28 14:09 backupfulltest
Apr 28 15:13 fullbackup
Core> █
```

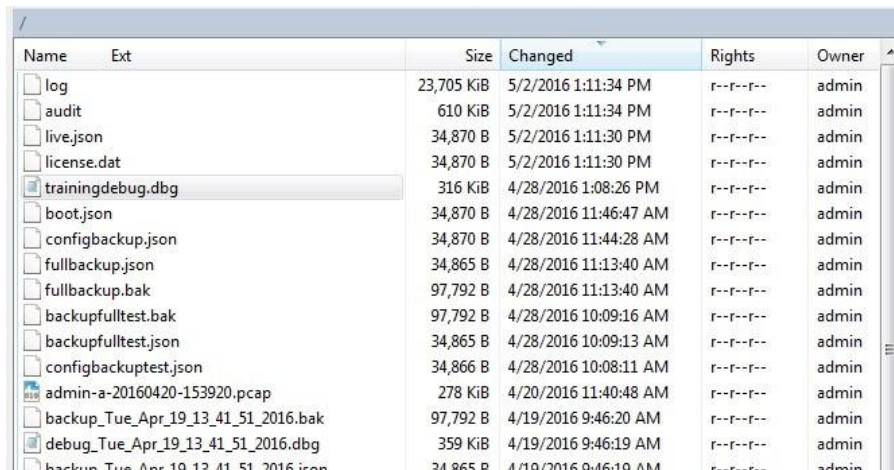
JSON and Backup Files

- live.json is the running mmp config on the server. This is what is copied when taking a config snapshot.
- The json contains all mmp configurations, but does not include API based configurations.
- The .bak file also contains this information, however it also includes the database info (which cannot be read).
- The .bak file cannot be edited even though some parts are readable.

```
{
  "system": {
    "modules": "1",
    "hostname": "Core",
    "timezone": "Etc\\UTC",
    "model": "VM",
    "version": "1.8.8"
  },
  "users": {
    "61646d696e": {
      "password": "$5$YvXmas2IsKNsmIEC$Wj\\5ivvDdp4wXriPDPXeiFkZrBhm2BEZDIJ4ddh10Y9",
      "salt": "$5$YvXmas2IsKNsmIEC$Wj\\5ivvDdp4wXriPDPXeiFkZrBhm2BEZDIJ4ddh10Y9",
      "reset": "false",
      "date": "2016-Oct-06",
      "lastchange": "2016-Apr-06",
      "uid": "0",
      "lock_time": "24332771",
      "failed_logins": "0",
      "last_login_time": "2016-Apr-28 15:13:11",
      "last_login": "Last login 2016-Apr-28 15:13:11 using SSH remote host 14.0.25.243"
    }
  },
  "ipv4": {
    "module": {
      "interfaces": {
        "eth4": {
          "dhcp": "false",
          "enabled": "true",
          "default": "true",
          "macaddress": "00:50:56:A1:1A:22",
          "address": "14.80.99.225",
          "prefixlen": "24",
          "gateway": "14.80.99.1"
        }
      }
    }
  }
}
```

Debug Files

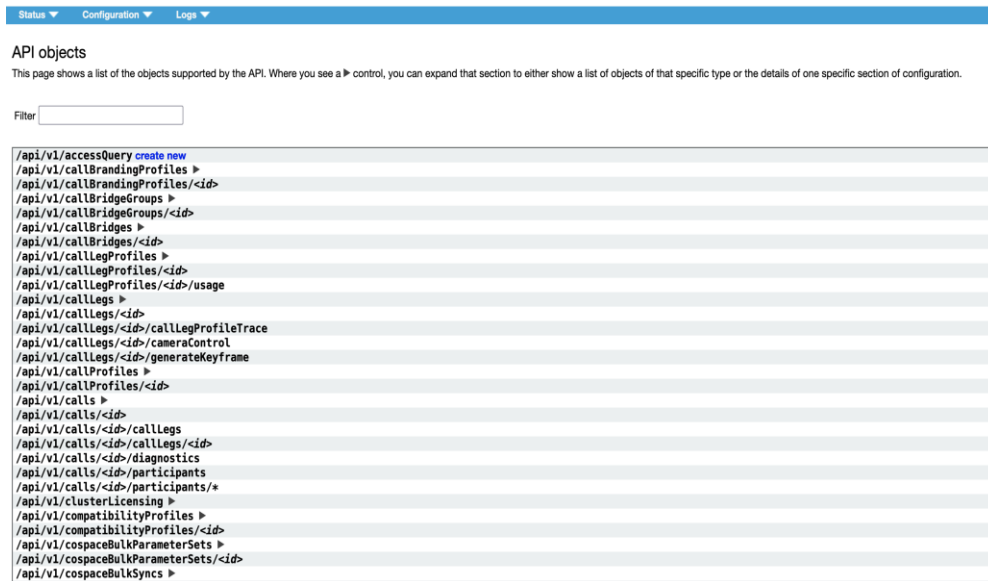
- Create with “*debug <filename>*” command.
- Can take some time to generate the file, and can be downloaded via winSCP from root directory.
- The debug file contains the following info...
 - File Structure
 - TCP/UDP Connections
 - ifconfig Output
 - Process List
 - CPU and Memory Usage
 - Kernel Logs
 - Disk Info
 - PostGRES Database Info
 - Logging Info



Name	Ext	Size	Changed	Rights	Owner
log		23,705 KiB	5/2/2016 1:11:34 PM	r--r--r--	admin
audit		610 KiB	5/2/2016 1:11:34 PM	r--r--r--	admin
live.json		34,870 B	5/2/2016 1:11:30 PM	r--r--r--	admin
license.dat		34,870 B	5/2/2016 1:11:30 PM	r--r--r--	admin
trainingdebug.dbg		316 KiB	4/28/2016 1:08:26 PM	r--r--r--	admin
boot.json		34,870 B	4/28/2016 11:46:47 AM	r--r--r--	admin
configbackup.json		34,870 B	4/28/2016 11:44:28 AM	r--r--r--	admin
fullbackup.json		34,865 B	4/28/2016 11:13:40 AM	r--r--r--	admin
fullbackup.bak		97,792 B	4/28/2016 11:13:40 AM	r--r--r--	admin
backupfulltest.bak		97,792 B	4/28/2016 10:09:16 AM	r--r--r--	admin
backupfulltest.json		34,865 B	4/28/2016 10:09:13 AM	r--r--r--	admin
configbackuptest.json		34,866 B	4/28/2016 10:08:11 AM	r--r--r--	admin
admin-a-20160420-153920.pcap		278 KiB	4/20/2016 11:40:48 AM	r--r--r--	admin
backup_Tue_Apr_19_13_41_51_2016.bak		97,792 B	4/19/2016 9:46:20 AM	r--r--r--	admin
debug_Tue_Apr_19_13_41_51_2016.dbg		359 KiB	4/19/2016 9:46:19 AM	r--r--r--	admin
backup_Tue_Apr_19_13_41_51_2016.json		34,865 B	4/19/2016 9:46:19 AM	r--r--r--	admin

API - GET

- Can be done with any browser
- Check Configuration/Status
- Returns XML
- Reference API guide



The screenshot shows a web interface with a blue header bar containing 'Status', 'Configuration', and 'Logs' tabs. Below the header, the title 'API objects' is displayed. A descriptive text states: 'This page shows a list of the objects supported by the API. Where you see a ► control, you can expand that section to either show a list of objects of that specific type or the details of one specific section of configuration.' Below this text is a 'Filter' input field. The main content area displays a list of API objects in XML format, each followed by a right-pointing arrow (►) indicating it is expandable. The objects listed are:

- /api/v1/accessQuery create new
- /api/v1/callBrandingProfiles ►
- /api/v1/callBrandingProfiles/<id>
- /api/v1/callBridgeGroups ►
- /api/v1/callBridgeGroups/<id>
- /api/v1/callBridges ►
- /api/v1/callBridges/<id>
- /api/v1/callLegProfiles ►
- /api/v1/callLegProfiles/<id>
- /api/v1/callLegProfiles/<id>/usage
- /api/v1/callLegs ►
- /api/v1/callLegs/<id>
- /api/v1/callLegs/<id>/callLegProfileTrace
- /api/v1/callLegs/<id>/cameraControl
- /api/v1/callLegs/<id>/generateKeyframe
- /api/v1/callProfiles ►
- /api/v1/callProfiles/<id>
- /api/v1/calls ►
- /api/v1/calls/<id>
- /api/v1/calls/<id>/callLegs
- /api/v1/calls/<id>/callLegs/<id>
- /api/v1/calls/<id>/diagnostics
- /api/v1/calls/<id>/participants
- /api/v1/calls/<id>/participants/*
- /api/v1/clusterlicensing ►
- /api/v1/compatibilityProfiles ►
- /api/v1/compatibilityProfiles/<id>
- /api/v1/cospaceBulkParameterSets ►
- /api/v1/cospaceBulkParameterSets/<id>
- /api/v1/cospaceBulkSyncs ►

API – Call Diagnostic Logs

- Require a POST – Extension required
- POST to /api/v1/coSpaces/<cospace-ID>/diagnostics
- Location Header in 200 OK gives Diagnostic ID
- GET to <location>/contents
- Diagnostics can also be seen in WebAdmin Status page

Response

POST on https://[redacted]:8443/api/v1/coSpaces/1a8929af-67f9-4f6a-bc25-da148c2ae29d/diagnostics
Status: 200 OK

Headers:

Date	Mon, 17 Apr 2017 18:34:48 GMT
Server	Apache
x-frame-options	DENY
Strict-Transport-Security	max-age=31536000; includeSubDomains
Location	/api/v1/system/diagnostics/86b35301-c285-45e8-9756-8503acaa910e
Vary	Accept-Encoding
Content-Encoding	gzip
Keep-Alive	timeout=5, max=100
Connection	Keep-Alive

Call diagnostic logs

☐	Log name	Time	Download link
☐	test 1.space	2017-04-17 18:34:49.009722 +0000	[download]
☐	test 1.space	2017-04-17 18:33:58.548672 +0000	[download]
☐	test 1.space	2017-04-17 18:30:33.464722 +0000	[download]
☐	test 1.space	2017-04-17 18:30:32.953933 +0000	[download]
☐	test 1.space	2017-04-17 18:30:32.387234 +0000	[download]
☐	test 1.space	2017-04-17 16:55:08.341660 +0000	[download]

Delete

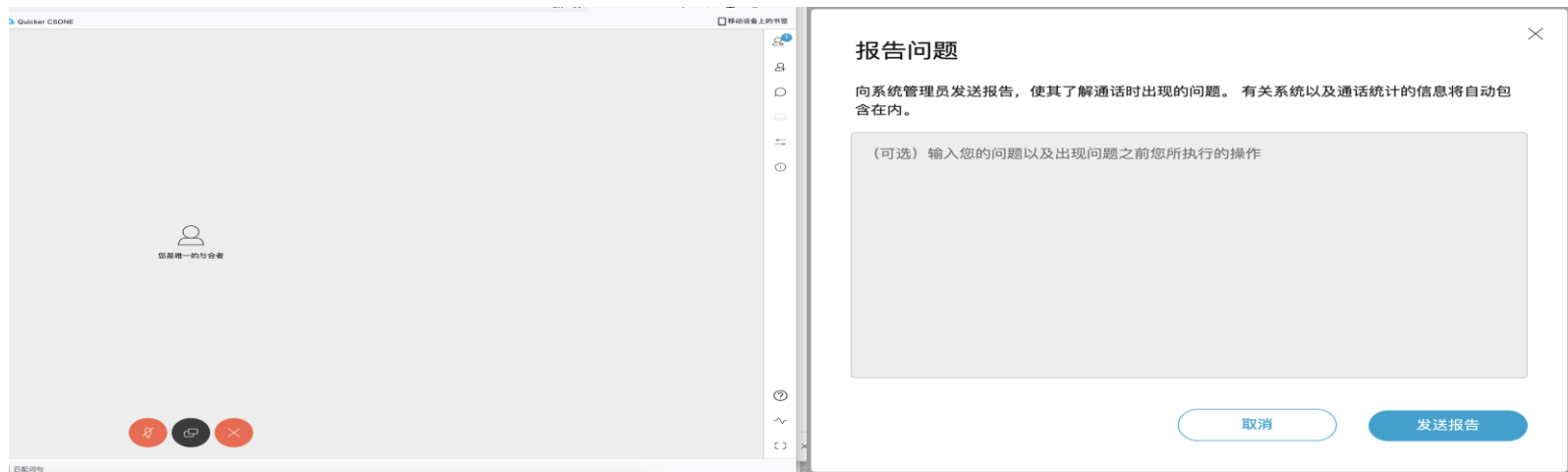
Delete all

CMA / WebRTC call Diagnostics

Client/Call Diagnostics

- If a problem is detected on a call using an CMA / WebRTC client, it may prompt the user to “send diagnostics”.
- These are saved locally on the device in the case of a PC or MAC from the clients point of view, and a diagnostic file is also created on the server running the callbridge it was connected to.
- This file can be seen and downloaded in the webadmin on the “general status” page
- The last 12 files are stored on the server.
- Includes information on the client software/hardware, relevant log lines from server, captured SDP and media info, and other diagnostics.
- Diagnostics can also be manually generated on the client by going to options -> diagnostics -> save.

Client/Call Diagnostics (Cont'd)



Client/Call Diagnostics (Cont'd)

System status

Uptime	24 days, 21 hours, 2 minutes
Build version	3.2.2
Media module status	1/1 (full media capacity)
Lync Edge registrations	not configured
web app calls	1
SIP calls	0
Lync calls	0
Forwarded calls	0
Completed calls	27
Activated conferences	1
Active Lync subscribers	0
Total outgoing media bandwidth	71.2 Kbit/s
Total incoming media bandwidth	1.06 Mbit/s

Fault conditions

Date	Time	Fault condition
none		

Recent errors and warnings

Date	Time	Logging level	Message
none			

Call diagnostic logs

☐	Log name	Time	Download link
☐	"guest1820157471" diagnostics	2022-05-23 06:58:35.019385 +0000	[download]

Client Diagnostic Log Examples

=== Model info =====

Unprocessed events 0

Num contacts: 130

Num calls: 28; in call list 28

Num cospaces: 28; in cospace list 28

Num endpoints: 10; in endpoint list 10

Num resources: 0; in resource list 0

Num recent calls: 3

Canary: 0x12345678

Audio mute status: unmuted

Video mute status: unmuted

Management server: callbridge1.acano.com

Control server: callbrdge1.acano.com

Server status <http://acano.com/management/callbridge1.acano.com@pubsub.acano.com>

Currently logged in clients:

8252fe0110076fb5 (this) : presence 3

Client Diagnostic Log Examples - Audio

```
2284 === Media sessions info =====
2285 ===== Media session FE48560 =====
2286   Change object awaiting 0 responses
2287   -- Message counter --
2288   Session offers sent 2 response 2, pending 0
2289   Session answers received 2
2290   Selections sent 2 response 2, pending 0
2291   Selections received 1
2292   Advertisement sent 1 response 1, pending 0
2293   Configures sent 0 response 0, pending 0
2294   Configures received 0
2295   Streams advertised:
2296   Audio: 25703630-d8ef-4e88-bbe9-8d3129d3016f
2297   Main video: de6ac16d-ec14-4fc1-9297-a83afc6bc6b7
2298   Streams requested from far end:
2299       bdc7d5b4-5a73-419b-8d09-ead2ff8c00ac with multiplexId 2 @ 1010x688; aspect ratio 1010:688; layout 3
2300       4e1a1422-0984-45a4-9577-dfc3c304d472 with multiplexId 1
2301   Audio session: 25703630-d8ef-4e88-bbe9-8d3129d3016f
2302   Remote media: 209.9.228.135:3478; control: 209.9.228.135:3478
2303       215 seconds ago: media -:50139; control: -:50391
2304       215 seconds ago: media 209.9.228.135:3478; control: 209.9.228.135:3478
2305       215 seconds ago: media 209.9.228.135:3478; control: 209.9.228.135:3478
2306   Estimated bandwidth: Unknown
2307   Interfaces:
2308       10.86.8.11 media 51164 control 51165
2309       ::1 media 51166 control 51167
2310       127.0.0.1 media 51168 control 51169
2311       2001::9d38:6ab8:3890:efe:8380:bbfb media 51170 control 51171
2312       fe80::3890:efe:8380:bbfb media 51172 control 51173
```

Audio session

Remote and local
media interface

Client Diagnostic Log Examples - Audio

```
2498 Ice complete: 1
2499 DTLS not started media complete 1 control complete 1
2500 Tx statistics:
2501     Multiplex ID      100
2502     SSRC              fa4104ef
2503     Payload type      98
2504     Codec              opus
2505     Clock rate        48000
2506     Packets Total     10569
2507     Bitrate           67.9 Kbps
2508     Round trip time   215ms
2509     Encryption        on
2510 Rx statistics:
2511     Multiplex ID      1
2512     SSRC              1cb7b15f
2513     Payload type      98
2514     Codec              opus
2515     Clock rate        48000
2516     Packets Total     10672
2517     Bitrate           77.5 Kbps
2518     Encryption        on
```



Chosen TX audio codec



Audio codec TX bitrate



Chosen RX audio codec



Audio codec RX bitrate

Client Diagnostic Log Examples - Video

```
2519 Video session: de6ac16d-ec14-4fc1-9297-a83afc6bc6b7
2520 Remote media: 209.9.228.135:3478; control: 209.9.228.135:3478
2521     215 seconds ago: media -:50389; control: -:50313
2522     215 seconds ago: media 209.9.228.135:3478; control: 209.9.228.135:3478
2523 Estimated bandwidth: 18454873 over 1 tx streams
2524 Interfaces:
2525     10.86.8.11 media 51174 control 51175
2526     ::1 media 51176 control 51177
2527     127.0.0.1 media 51178 control 51179
2528     2001::9d38:6ab8:3890:efe:8380:bbfb media 51180 control 51181
2529     fe80::3890:efe:8380:bbfb media 51182 control 51183
```

Remote and local
media interface

Client Diagnostic Log Examples - Video

```
2715 Ice complete: 1
2716 DTLS not started media complete 1 control complete 1
2717 Tx statistics:
2718     Multiplex ID      200
2719     SSRC              e7ee89ea
2720     Payload type      96
2721     Codec              H264
2722     Clock rate         90000
2723     Packets Total      17619
2724     Dropped 2 Curr 0.0 pct
2725     Bitrate            720.4 Kbps
2726     Round trip time    217ms
2727     Encryption         on
2728     Resolution         768x448
2729     Framerate          28.9 fps
2730     Keyframes Requests 6
2731     Keyframes          0
2732     Configured bitrate 1024.0 Kbps
2733     Flow control bitrate 1024.0 Kbps
```

Chosen TX video codec

0.0% packet loss TX
Video codec TX bitrate

keyframe requests TX

Client Diagnostic Log Examples - Video

```
2734 Rx statistics:
2735     Multiplex ID      2
2736     SSRC              5501d8bc
2737     Payload type      96
2738     Codec             H264
2739     Clock rate        90000
2740     Packets Total     18526
2741         FEC 9
2742         Dropped 72 Curr 0.0 pct
2743     Bitrate           721.6 Kbps
2744     Encryption       on
2745     Resolution        768x448
2746     Framerate         25.3 fps
2747     Keyframes Requests 363
2748     Keyframes         59
2749     Configured bitrate 0.0 Kbps
2750     Flow control bitrate 0.0 Kbps
```



Chosen RX video codec



0.0% packet loss RX
Video codec RX bitrate



keyframe requests RX

Chrome WebRTC debugging

- chrome://webrtc-internals/

Chrome | chrome://webrtc-internals

► Create Dump

Read Stats From:

<https://10.106.102.202/en-us/meeting/661000> [61278-1]

<https://10.106.102.202/en-us/meeting/661000>, { iceServers: [], iceTransportPolicy: all, bundlePolicy: balanced, true}}}

Time	Event
14/12/2020, 16:28:32	► transceiverAdded
14/12/2020, 16:28:32	► transceiverAdded
14/12/2020, 16:28:32	negotiationneeded
14/12/2020, 16:28:32	► createOffer
14/12/2020, 16:28:32	► createOfferOnSuccess
14/12/2020, 16:28:32	► setLocalDescription (munged)
14/12/2020, 16:28:32	setLocalDescriptionOnSuccess
14/12/2020, 16:28:32	► signalingstatechange
14/12/2020, 16:28:32	► transceiverModified
14/12/2020, 16:28:32	► transceiverModified
14/12/2020, 16:28:32	► transceiverModified
14/12/2020, 16:28:32	► icegatheringstatechange
14/12/2020, 16:28:32	► icecandidate (host)
14/12/2020, 16:28:32	► icecandidate (host)
14/12/2020, 16:28:32	► icecandidate (host)
14/12/2020, 16:28:32	► icecandidate (host)
14/12/2020, 16:28:32	► icecandidate (host)
14/12/2020, 16:28:32	► icecandidate (host)
14/12/2020, 16:28:32	► icecandidate (host)
14/12/2020, 16:28:32	► icecandidate (host)
14/12/2020, 16:28:32	► icecandidate (host)
14/12/2020, 16:28:32	► iceconnectionstatechange
14/12/2020, 16:28:32	► iceconnectionstatechange (legacy)
14/12/2020, 16:28:32	► connectionstatechange
14/12/2020, 16:28:32	► setRemoteDescription
14/12/2020, 16:28:32	► icecandidate (host)

Mozilla WebRTC debugging

- `about:webrtc`

Save Page Start Debug Mode Start AEC Logging

Session Statistics

Clear History

[948 | 674309865479] <https://10.106.102.202/en-us/meeting/661000?trace=true> Mon Dec 14 2020 16:25:53 GMT+0530 (India Standard Time)

Show tab

▲ hide details

PeerConnection ID: 1607943351036196 (id=674309865479 url=<https://10.106.102.202/en-us/meeting/661000?trace=true>)

RTCCConfiguration
bundlePolicy: balanced
iceTransportPolicy: all
peerIdentity(Provided/Not Provided): Not Provided
sdpSemantics: Not Provided
iceServers:

ICE Stats

Trickled candidates (arriving after answer) are highlighted in blue

ICE State	Nominated	Selected	Local Candidate	Remote Candidate	Component ID	Priority	Bytes sent	Bytes received
succeeded	true	true	10.65.34.50:57088/udp(host) [non-proxied]	10.106.102.202:58619/udp(host)	1	9114442320345825000	21152251	825677
succeeded	true	true	10.65.34.50:57088/udp(host) [non-proxied]	10.106.102.202:58619/udp(host)	1	9114442320345825000	21152251	825677
succeeded	true	true	10.65.34.50:57088/udp(host) [non-proxied]	10.106.102.202:58619/udp(host)	1	9114442320345825000	21152251	825677
succeeded	true	true	10.65.34.50:57088/udp(host) [non-proxied]	10.106.102.202:58619/udp(host)	1	9114442320345825000	21152251	825677
failed	false	false	192.168.1.6:51755/udp(host) [non-proxied]	10.106.102.202:58619/udp(host)	1	9115005270299247000	0	0
failed	false	false	192.168.1.6:51755/udp(host) [non-proxied]	10.106.102.202:58619/udp(host)	1	9115005270299247000	0	0
failed	false	false	192.168.1.6:51755/udp(host) [non-proxied]	10.106.102.202:58619/udp(host)	1	9115005270299247000	0	0
failed	false	false	192.168.1.6:51755/udp(host) [non-proxied]	10.106.102.202:58619/udp(host)	1	9115005270299247000	0	0

CMA / CMS Server Crash

CMS Crash Dump Files

 Module-0-Crash-37dbbd3f-1781-a49f-412cc7a8-77012c9c	Crash Dump File	3,014 KB
 Module-0-Crash-1663c2a8-4ef4-14cd-28171e93-7fc95dd9	Crash Dump File	1,829 KB
 Module-1-Crash-37d33784-1fe0-e011-3cb6f17c-13c0217e	Crash Dump File	1,331 KB
 Module-5-Crash-37d33784-1fe0-e011-3cb6f17c-13c0217e	Crash Dump File	1,341 KB
 Module-9-Crash-37d33784-1fe0-e011-3cb6f17c-13c0217e	Crash Dump File	1,365 KB

- In the event that a module crashes, a crash dump file will be created on the server.
- You can view and download these files via winSCP.
- The file name will indicate the module that has crashed.
 - Module 0 generally indicates a full system crash resulting in a reboot.
 - Other modules usually indicate a DSP issue and can result in a quick black screen on video, but should not bring down the server.

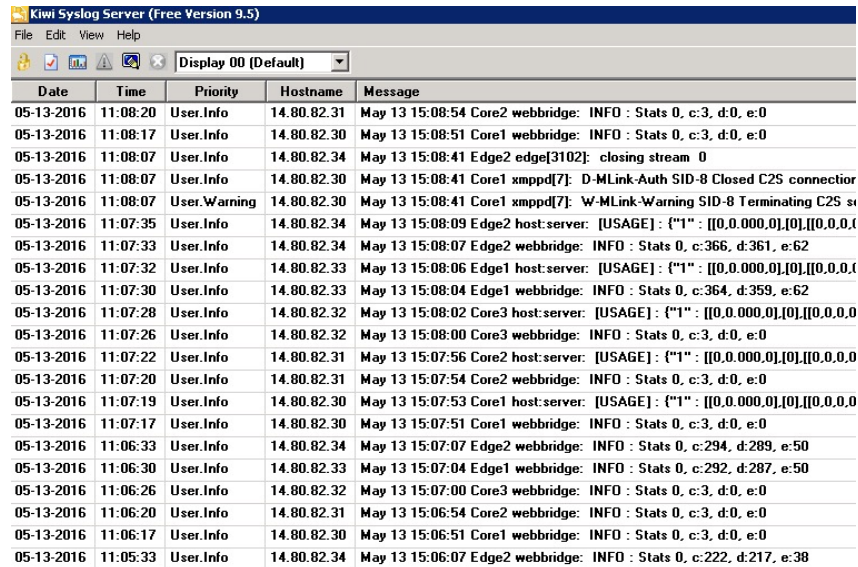
CMA Crash Dump Files

- PC client crash dump files are also collected and stored on the local machine.
- They can be found in
C:\Users\<username>\AppData\Roaming\Acano\Client

Name	Date modified	Type	Size
 acanodump_20130718_100147_coredump	18/07/2013 11:01	Crash Dump File	173 KB
 acanodump_20130730_073837_coredump	30/07/2013 08:38	Crash Dump File	163 KB
 acanodump_20130730_081847_coredump	30/07/2013 09:18	Crash Dump File	182 KB

Syslog Server

- With a multi-server deployment, it is very important to configure syslog monitoring to have a single source of logs from all servers, rather than having to manually sftp to each and download the log files.
- To configure syslog on the CMS servers, use the following...
 - “syslog server add <IP/hostname> <port>”
 - “syslog enable”
- CMS servers use TCP for syslog output.
- Syslog can also be set to output the audit logs.
- Debugs enabled via the webadmin page will be output to syslog.



The screenshot shows the Kiwi Syslog Server interface with a table of received messages. The table has columns for Date, Time, Priority, Hostname, and Message. The messages are timestamped May 13 15:08 and include details about Core2 and Edge2 webbridges, Core1 and Edge2 host servers, and Core1 and Edge2 xmppd processes.

Date	Time	Priority	Hostname	Message
05-13-2016	11:08:20	User.Info	14.80.82.31	May 13 15:08:54 Core2 webbridge: INFO : Stats 0, c:3, d:0, e:0
05-13-2016	11:08:17	User.Info	14.80.82.30	May 13 15:08:51 Core1 webbridge: INFO : Stats 0, c:3, d:0, e:0
05-13-2016	11:08:07	User.Info	14.80.82.34	May 13 15:08:41 Edge2 edge[3102]: closing stream 0
05-13-2016	11:08:07	User.Info	14.80.82.30	May 13 15:08:41 Core1 xmppd[7]: D-MLink-Auth SID-8 Closed C2S connector
05-13-2016	11:08:07	User.Warning	14.80.82.30	May 13 15:08:41 Core1 xmppd[7]: W-MLink-Warning SID-8 Terminating C2S s
05-13-2016	11:07:35	User.Info	14.80.82.34	May 13 15:08:09 Edge2 host:server: [USAGE] : {"1" : [[0.0.000.0],[0],[0.0.0.0]
05-13-2016	11:07:33	User.Info	14.80.82.34	May 13 15:08:07 Edge2 webbridge: INFO : Stats 0, c:366, d:361, e:62
05-13-2016	11:07:32	User.Info	14.80.82.33	May 13 15:08:06 Edge1 host:server: [USAGE] : {"1" : [[0.0.000.0],[0],[0.0.0.0]
05-13-2016	11:07:30	User.Info	14.80.82.33	May 13 15:08:04 Edge1 webbridge: INFO : Stats 0, c:364, d:359, e:62
05-13-2016	11:07:28	User.Info	14.80.82.32	May 13 15:08:02 Core3 host:server: [USAGE] : {"1" : [[0.0.000.0],[0],[0.0.0.0]
05-13-2016	11:07:26	User.Info	14.80.82.32	May 13 15:08:00 Core3 webbridge: INFO : Stats 0, c:3, d:0, e:0
05-13-2016	11:07:22	User.Info	14.80.82.31	May 13 15:07:56 Core2 host:server: [USAGE] : {"1" : [[0.0.000.0],[0],[0.0.0.0]
05-13-2016	11:07:20	User.Info	14.80.82.31	May 13 15:07:54 Core2 webbridge: INFO : Stats 0, c:3, d:0, e:0
05-13-2016	11:07:19	User.Info	14.80.82.30	May 13 15:07:53 Core1 host:server: [USAGE] : {"1" : [[0.0.000.0],[0],[0.0.0.0]
05-13-2016	11:07:17	User.Info	14.80.82.30	May 13 15:07:51 Core1 webbridge: INFO : Stats 0, c:3, d:0, e:0
05-13-2016	11:06:33	User.Info	14.80.82.34	May 13 15:07:07 Edge2 webbridge: INFO : Stats 0, c:294, d:289, e:50
05-13-2016	11:06:30	User.Info	14.80.82.33	May 13 15:07:04 Edge1 webbridge: INFO : Stats 0, c:292, d:287, e:50
05-13-2016	11:06:26	User.Info	14.80.82.32	May 13 15:07:00 Core3 webbridge: INFO : Stats 0, c:3, d:0, e:0
05-13-2016	11:06:20	User.Info	14.80.82.31	May 13 15:06:54 Core2 webbridge: INFO : Stats 0, c:3, d:0, e:0
05-13-2016	11:06:17	User.Info	14.80.82.30	May 13 15:06:51 Core1 webbridge: INFO : Stats 0, c:3, d:0, e:0
05-13-2016	11:05:33	User.Info	14.80.82.34	May 13 15:06:07 Edge2 webbridge: INFO : Stats 0, c:222, d:217, e:38

Troubleshooting Clustering

Database Clustering – Best practice

- Do not forget to issue “**database cluster upgrade_schema**” command on Master after upgrading the Database cluster
- Minimum 3 servers are required for Database clustering
- Restoring Older version backup to newer version is possible but NOT newer version backup to older version
- All nodes check the status of each other at regular intervals.
- If one of these checks fails or takes too long to complete, that node will be considered disconnected.
- If five of the checks fail in a row, a new master will be elected and all slave nodes will now reference the master

Mar 14 12:43:40 example_server01 sfpool: Health check 115.114.248.70 master check failure:

Mar 14 12:43:41 example_server01 sfpool: health check: up check to 115.114.248.70:5432 failed with telnet: can't connect to remote host (115.114.248.70): Connection refused#012telneterr

Mar 14 12:43:49 example_server01 sfpool: Failover monitor: Failing over from node 115.114.248.70 to node 115.114.249.70

Database Clustering

```
acano> database cluster status
Status                               : Enabled

Nodes:
  10.124.56.210 (me)                 : Connected Primary
  10.124.56.211                       : Connected Replica ( 56 bytes behind )
  10.124.56.212                       : Connected Replica ( In Sync )
Node in use                          : 10.124.56.210

Interface                            : a

Certificates
  Server Key                         : dbs.key
  Server Certificate                  : dbs.cer
  Client Key                         : dbc.key
  Client Certificate                  : dbc.cer
  CA Certificate                      : root.cer

Last command                         : 'database cluster upgrade_schema' (Success)

acano> █
```

CallBridge Clustering – Best practice

Call Bridge identity

Status ▼ Configuration ▼ Logs ▼ User: admin ▼

Call Bridge identity

Unique name

Peer link bit rate

Participant limit

Clustered Call Bridges

☐	Unique name	Address	Peer link SIP domain	Status	
☐	CB_cms02	https://cms02.cisco.lab:8443		connection active; time since last heartbeat: 3 seconds	[edit]
☐	CB_cms01	https://cms01.cisco.lab:8443		[this Call Bridge]	[edit]
☐	CB_cms03	https://cms03.cisco.lab:8443		connection active; time since last heartbeat: 2 seconds	[edit]
					Add New Reset

Questions?